YERRAGUNTLA KOUSHIK

23BTRCM014

JAIN(Deemed-to-be-University)

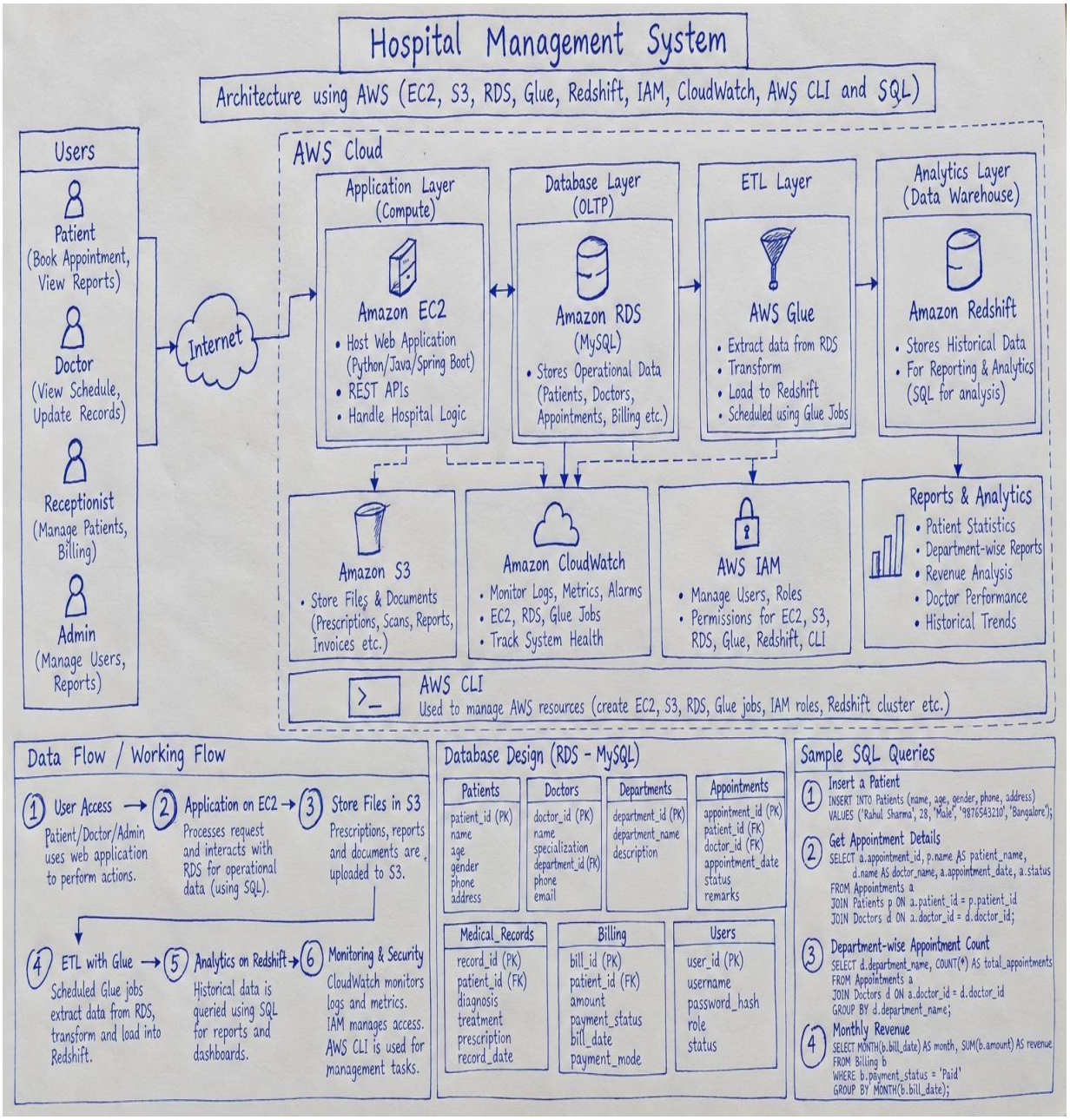
CLOUD TECHNOLOGY AND MOBILE APPLICATION

2023 - 2027

Date : 10/10/2026

TITLE : AWS Cloud & Data Engineering




Hospital Management System
Architecture using AWS (EC2, S3, RDS, Glue, Redshift, IAM, CloudWatch, AWS CLI and SQL)
Users
Patient
(Book Appointment, View Reports)
Doctor
(View Schedule, Update Records)
Receptionist
(Manage Patients, Billing)
Admin
(Manage Users, Reports)
Internet
AWS Cloud
Application Layer (Compute)
Amazon EC2
• Host Web Application (Python/Java/Spring Boot)
• REST APIs
• Handle Hospital Logic
Database Layer (OLTP)
Amazon RDS
(MySQL)
• Stores Operational Data (Patients, Doctors, Appointments, Billing etc.)
ETL Layer
AWS Glue
• Extract data from RDS
• Transform
• Load to Redshift
• Scheduled using Glue Jobs
Analytics Layer (Data Warehouse)
Amazon Redshift
• Stores Historical Data
• For Reporting & Analytics (SQL for analysis)
Amazon S3
• Store Files & Documents (Prescriptions, Scans, Reports, Invoices etc.)
Amazon CloudWatch
• Monitor Logs, Metrics, Alarms
• EC2, RDS, Glue Jobs
• Track System Health
AWS IAM
• Manage Users, Roles
• Permissions for EC2, S3, RDS, Glue, Redshift, CLI
Reports & Analytics
• Patient Statistics
• Department-wise Reports
• Revenue Analysis
• Doctor Performance
• Historical Trends
AWS CLI
Used to manage AWS resources (create EC2, S3, RDS, Glue jobs, IAM roles, Redshift cluster etc.)
Data Flow / Working Flow
1 User Access → Patient/Doctor/Admin uses web application to perform actions.
2 Application on EC2 → Processes request and interacts with RDS for operational data (using SQL).
3 Store Files in S3 → Prescriptions, reports and documents are uploaded to S3.
4 ETL with Glue → Scheduled Glue jobs extract data from RDS, transform and load into Redshift.
5 Analytics on Redshift → Historical data is queried using SQL for reports and dashboards.
6 Monitoring & Security → CloudWatch monitors logs and metrics. IAM manages access. AWS CLI is used for management tasks.
Database Design (RDS - MySQL)
Patients
patient_id (PK)
name
age
gender
phone
address
Doctors
doctor_id (PK)
name
specialization
department_id (FK)
phone
email
Departments
department_id (PK)
department_name
description
Appointments
appointment_id (PK)
patient_id (FK)
doctor_id (FK)
appointment_date
status
remarks
Medical_Records
record_id (PK)
patient_id (FK)
diagnosis
treatment
prescription
record_date
Billing
bill_id (PK)
patient_id (FK)
amount
payment_status
bill_date
payment_mode
Users
user_id (PK)
username
password_hash
role
status
Sample SQL Queries
1 Insert a Patient
INSERT INTO Patients (name, age, gender, phone, address)
VALUES ('Rahul Sharma', 28, 'Male', '9876543210', 'Bangalore');
2 Get Appointment Details
SELECT a.appointment_id, p.name AS patient_name,
d.name AS doctor_name, a.appointment_date, a.status
FROM Appointments a
JOIN Patients p ON a.patient_id = p.patient_id
JOIN Doctors d ON a.doctor_id = d.doctor_id;
3 Department-wise Appointment Count
SELECT d.department_name, COUNT(*) AS total_appointments
FROM Appointments a
JOIN Doctors d ON a.doctor_id = d.doctor_id
GROUP BY d.department_name;
4 Monthly Revenue
SELECT MONTH(b.bill_date) AS month, SUM(b.amount) AS revenue
FROM Billing b
WHERE b.payment_status = 'Paid'
GROUP BY MONTH(b.bill_date);

Hospital Management System Using SQL

```sql
create database hospitaldb;
use hospitaldb;

-- PATIENTS TABLE --
CREATE TABLE Patients (
    patient_id INT PRIMARY KEY,
    patient_name VARCHAR(50),
    age INT,
    gender VARCHAR(10),
    city VARCHAR(50),
    admission_date DATE,
    doctor_id INT
);
```

```sql
INSERT INTO Patients VALUES
(1, 'Aarav', 25, 'Male', 'Bengaluru', '2026-01-10', 101),
(2, 'Priya', 32, 'Female', 'Hyderabad', '2026-01-15', 102),
(3, 'Rahul', 45, 'Male', 'Chennai', '2026-02-05', 103),
(4, 'Sneha', 28, 'Female', 'Mumbai', '2026-02-12', 101),
(5, 'Arjun', 60, 'Male', 'Pune', '2026-03-01', 104),
(6, 'Divya', 38, 'Female', 'Bengaluru', '2026-03-10', 102),
(7, 'Kiran', 52, 'Male', 'Mysuru', '2026-04-08', 105),
(8, 'Ananya', 19, 'Female', 'Chennai', '2026-04-20', 103),
(9, 'Vikram', 67, 'Male', 'Hyderabad', '2026-05-05', 104),
(10, 'Meera', 41, 'Female', 'Mumbai', '2026-05-18', 105);

select * from Patients;
```

Result Grid

patient_id	patient_name	age	gender	city	admission_date	doctor_id
1	Aarav	25	Male	Bengaluru	2026-01-10	101
2	Priya	32	Female	Hyderabad	2026-01-15	102
3	Rahul	45	Male	Chennai	2026-02-05	103
4	Sneha	28	Female	Mumbai	2026-02-12	101
5	Arjun	60	Male	Pune	2026-03-01	104
6	Divya	38	Female	Bengaluru	2026-03-10	102
7	Kiran	52	Male	Mysuru	2026-04-08	105
8	Ananya	19	Female	Chennai	2026-04-20	103
9	Vikram	67	Male	Hyderabad	2026-05-05	104
10	Meera	41	Female	Mumbai	2026-05-18	105
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Patients 5

```sql
-- DOCTOR TABLE --
CREATE TABLE Doctors (
    doctor_id INT PRIMARY KEY,
    doctor_name VARCHAR(50),
    specialization VARCHAR(50),
    experience INT,
    consultation_fee DECIMAL(10,2)
);

INSERT INTO Doctors VALUES
(101, 'Dr. Anil Kumar', 'Cardiology', 12, 800.00),
(102, 'Dr. Priya Sharma', 'Dermatology', 8, 600.00),
(103, 'Dr. Ravi Patel', 'Neurology', 15, 1200.00),
(104, 'Dr. Sneha Reddy', 'Orthopedics', 10, 900.00),
(105, 'Dr. Kiran Rao', 'Pediatrics', 7, 500.00),
(106, 'Dr. Meera Nair', 'Gynecology', 11, 1000.00),
(107, 'Dr. Arjun Singh', 'Cardiology', 6, 700.00),
(108, 'Dr. Divya Menon', 'Dermatology', 5, 450.00),
(109, 'Dr. Vikram Das', 'Neurology', 9, 950.00),
(110, 'Dr. Ananya Gupta', 'Orthopedics', 14, 1100.00);

Select * from Doctors;
```

```sql
Select * from Doctors;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

doctor_id	doctor_name	specialization	experience	consultation_fee
101	Dr. Anil Kumar	Cardiology	12	800.00
102	Dr. Priya Sharma	Dermatology	8	600.00
103	Dr. Ravi Patel	Neurology	15	1200.00
104	Dr. Sneha Reddy	Orthopedics	10	900.00
105	Dr. Kiran Rao	Pediatrics	7	500.00
106	Dr. Meera Nair	Gynecology	11	1000.00
107	Dr. Arjun Singh	Cardiology	6	700.00
108	Dr. Divya Menon	Dermatology	5	450.00
109	Dr. Vikram Das	Neurology	9	950.00
110	Dr. Ananya Gupta	Orthopedics	14	1100.00
NULL	NULL	NULL	NULL	NULL

Doctors 6

```sql

-- BILLING TABLE --
CREATE TABLE Billing (
    bill_id INT PRIMARY KEY,
    patient_id INT,
    bill_date DATE,
    amount DECIMAL(10,2),
    payment_status VARCHAR(20),
    payment_method VARCHAR(20),
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id)
);

INSERT INTO Billing VALUES
(1, 1, '2026-01-12', 5000.00, 'Paid', 'UPI'),
(2, 2, '2026-01-18', 7500.00, 'Pending', 'Card'),
(3, 3, '2026-02-08', 12000.00, 'Paid', 'Cash'),
(4, 4, '2026-02-15', 6500.00, 'Pending', 'UPI'),
(5, 5, '2026-03-05', 15000.00, 'Paid', 'Insurance'),
(6, 6, '2026-03-12', 8200.00, 'Paid', 'Card'),
(7, 7, '2026-04-10', 9500.00, 'Pending', 'Insurance'),
(8, 8, '2026-04-22', 4000.00, 'Paid', 'Cash'),
(9, 9, '2026-05-08', 18000.00, 'Pending', 'Card'),
(10, 10, '2026-05-20', 6800.00, 'Paid', 'UPI');

Select * from Billing;
```

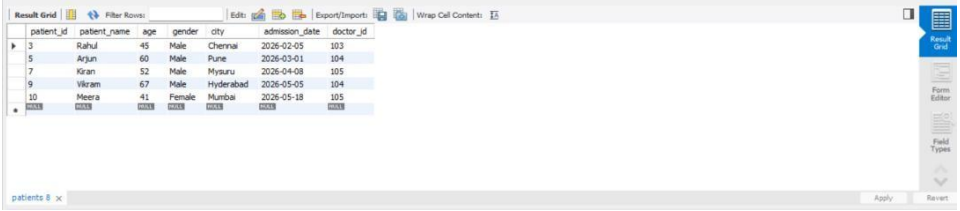
```sql
Select * from Billing;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

bill_id	patient_id	bill_date	amount	payment_status	payment_method
1	1	2026-01-12	5000.00	Paid	UPI
2	2	2026-01-18	7500.00	Pending	Card
3	3	2026-02-08	12000.00	Paid	Cash
4	4	2026-02-15	6500.00	Pending	UPI
5	5	2026-03-05	15000.00	Paid	Insurance
6	6	2026-03-12	8200.00	Paid	Card
7	7	2026-04-10	9500.00	Pending	Insurance
8	8	2026-04-22	4000.00	Paid	Cash
9	9	2026-05-08	18000.00	Pending	Card
10	10	2026-05-20	6800.00	Paid	UPI
NULL	NULL	NULL	NULL	NULL	NULL

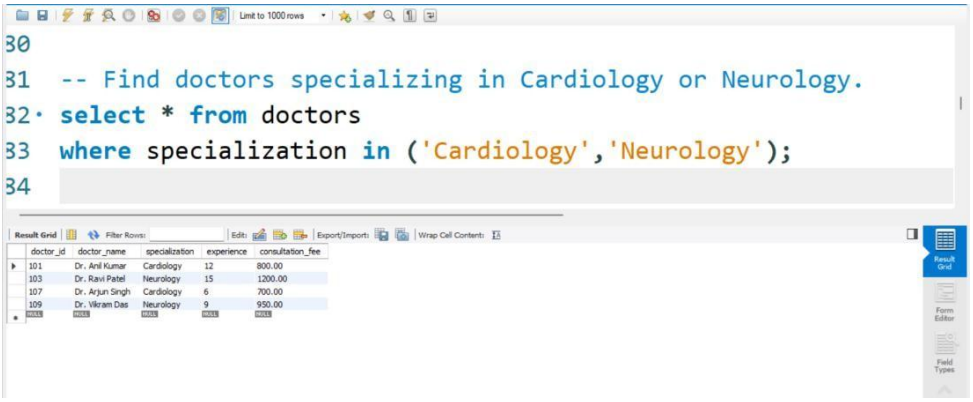
Billing 7 ×

```sql
-- Display patient details for patients older than 40.--
select * from patients
where age > 40;
```



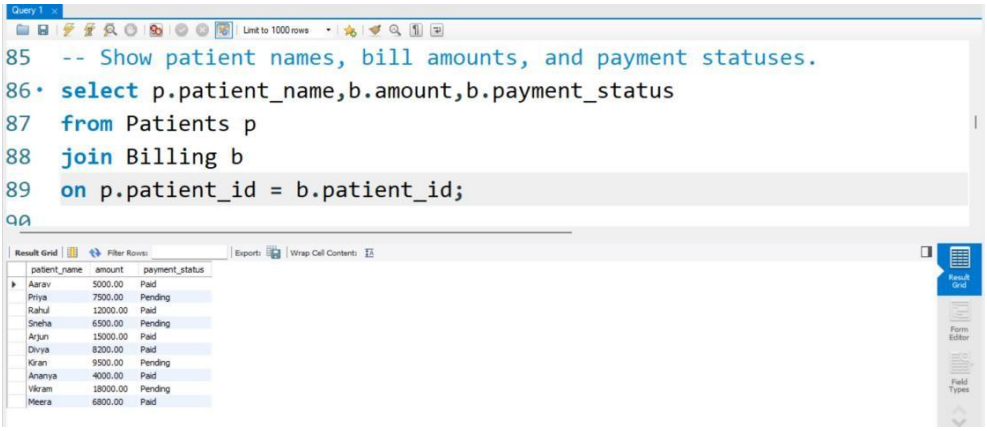
patient_id	patient_name	age	gender	city	admission_date	doctor_id
3	Rahul	45	Male	Chennai	2026-02-05	103
5	Arjun	60	Male	Pune	2026-03-01	104
7	Kiran	52	Male	Mysuru	2026-04-08	105
9	Vikram	67	Male	Hyderabad	2026-05-05	104
10	Meera	41	Female	Mumbai	2026-05-18	105

```sql
-- Find doctors specializing in Cardiology or Neurology.
select * from doctors
where specialization in ('Cardiology','Neurology');
```



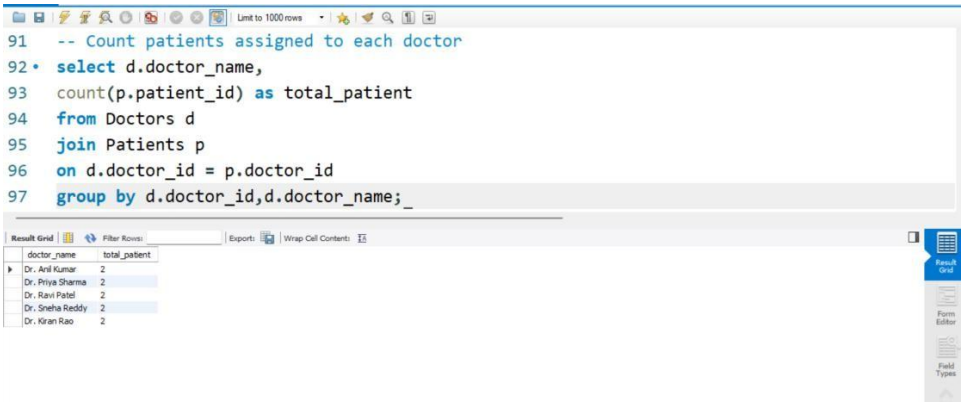
doctor_id	doctor_name	specialization	experience	consultation_fee
101	Dr. Anil Kumar	Cardiology	12	800.00
103	Dr. Ravi Patel	Neurology	15	1200.00
107	Dr. Arjun Singh	Cardiology	6	700.00
109	Dr. Vikram Das	Neurology	9	950.00

```sql
-- Show patient names, bill amounts, and payment statuses.
select p.patient_name,b.amount,b.payment_status
from Patients p
join Billing b
on p.patient_id = b.patient_id;
```



patient_name	amount	payment_status
Aarav	5000.00	Paid
Priya	7500.00	Pending
Rahul	12000.00	Paid
Sneha	6500.00	Pending
Arjun	15000.00	Paid
Divya	8200.00	Paid
Kiran	9500.00	Pending
Ananya	4000.00	Paid
Vikram	18000.00	Pending
Meera	6800.00	Paid

```sql
-- Count patients assigned to each doctor
select d.doctor_name,
count(p.patient_id) as total_patient
from Doctors d
join Patients p
on d.doctor_id = p.doctor_id
group by d.doctor_id,d.doctor_name;
```



doctor_name	total_patient
Dr. Anil Kumar	2
Dr. Priya Sharma	2
Dr. Ravi Patel	2
Dr. Sneha Reddy	2
Dr. Kiran Rao	2

```sql
-- Find doctors with more than one patient
SELECT d.doctor_name,COUNT(p.patient_id) AS total_patients
FROM Doctors d
JOIN Patients p
ON d.doctor_id = p.doctor_id
GROUP BY d.doctor_id, d.doctor_name
HAVING COUNT(p.patient_id) > 1;
```

doctor_name	total_patients
Dr. Anil Kumar	2
Dr. Priya Sharma	2
Dr. Ravi Patel	2
Dr. Sneha Reddy	2
Dr. Kiran Rao	2

```sql
-- Find bills above the average bill amount
SELECT p.patient_name, b.amount
FROM Patients p
JOIN Billing b
ON p.patient_id = b.patient_id
WHERE b.amount > (
SELECT AVG(amount)
    FROM Billing);
```

patient_name	amount
Rahul	12000.00
Arjun	15000.00
Kiran	9500.00
Vikram	18000.00

```sql
-- Calculate revenue by doctor
SELECT d.doctor_name, SUM(b.amount) AS total_revenue
FROM Doctors d
JOIN Patients p
ON d.doctor_id = p.doctor_id
JOIN Billing b
ON p.patient_id = b.patient_id
GROUP BY d.doctor_id, d.doctor_name
ORDER BY total_revenue DESC;
```

doctor_name	total_revenue
Dr. Sneha Reddy	33000.00
Dr. Kiran Rao	16300.00
Dr. Ravi Patel	16000.00
Dr. Priya Sharma	15700.00
Dr. Anil Kumar	11500.00

```sql
-- Find patients with high bills and pending payments
SELECT p.patient_name, p.city, b.amount, b.payment_status
FROM Patients p
JOIN Billing b
ON p.patient_id = b.patient_id
WHERE b.amount > 10000
AND b.payment_status = 'Pending';
```

patient_name	city	amount	payment_status
Vikram	Hyderabad	18000.00	Pending

```sql
-- Find patients whose bills are above ₹15,000 using a CTE
WITH HighBills AS (
    SELECT patient_id, amount
    FROM Billing
    WHERE amount > 15000
)
SELECT p.patient_name, hb.amount
FROM HighBills hb
JOIN Patients p
ON hb.patient_id = p.patient_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

patient_name	amount
Vikram	18000.00

```sql
-- Classify hospital bills into categories based on their amount using a CASE statement.
SELECT bill_id, amount,
    CASE
        WHEN amount >= 20000 THEN 'High'
        WHEN amount >= 10000 THEN 'Medium'
        ELSE 'Low'
    END AS bill_category
FROM Billing;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

bill_id	amount	bill_category
1	5000.00	Low
2	7500.00	Low
3	12000.00	Medium
4	6500.00	Low
5	15000.00	Medium
6	8200.00	Low
7	9500.00	Low
8	4000.00	Low
9	18000.00	Medium
10	6800.00	Low

```sql
-- Rank doctors by number of patients
WITH DoctorCounts AS (
    SELECT d.doctor_id, d.doctor_name, COUNT(p.patient_id) AS total_patients
    FROM Doctors d
    LEFT JOIN Patients p
        ON d.doctor_id = p.doctor_id
    GROUP BY d.doctor_id, d.doctor_name
)
SELECT doctor_name, total_patients,
RANK() OVER (
        ORDER BY total_patients DESC) AS doctor_rank
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

doctor_name	total_patients	doctor_rank
Dr. Anil Kumar	2	1
Dr. Priya Sharma	2	1
Dr. Ravi Patel	2	1
Dr. Sneha Reddy	2	1
Dr. Kiran Rao	2	1
Dr. Meera Nair	0	6
Dr. Arjun Singh	0	6
Dr. Divya Menon	0	6
Dr. Vikram Das	0	6
Dr. Ananya Gupta	0	6

```sql
-- Compare each bill with the previous bill
SELECT bill_id, bill_date, amount,
LAG(amount) OVER (
        ORDER BY bill_date, bill_id
    ) AS previous_bill,
    amount - LAG(amount) OVER (
        ORDER BY bill_date, bill_id
    ) AS bill_difference
FROM Billing;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

bill_id	bill_date	amount	previous_bill	bill_difference
1	2026-01-12	5000.00	NULL	NULL
2	2026-01-18	7500.00	5000.00	2500.00
3	2026-02-08	12000.00	7500.00	4500.00
4	2026-02-15	6500.00	12000.00	-5500.00
5	2026-03-05	15000.00	6500.00	8500.00
6	2026-03-12	8200.00	15000.00	-6800.00
7	2026-04-10	9500.00	8200.00	1300.00
8	2026-04-22	4000.00	9500.00	-5500.00
9	2026-05-08	18000.00	4000.00	14000.00
10	2026-05-20	6800.00	18000.00	-11200.00

```sql
-- Find the next bill amount
SELECT bill_id,bill_date, amount,
LEAD(amount) OVER (
        ORDER BY bill_date, bill_id
    ) AS next_bill_amount
FROM Billing;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

bill_id	bill_date	amount	next_bill_amount
1	2026-01-12	5000.00	7500.00
2	2026-01-18	7500.00	12000.00
3	2026-02-08	12000.00	6500.00
4	2026-02-15	6500.00	15000.00
5	2026-03-05	15000.00	8200.00
6	2026-03-12	8200.00	9500.00
7	2026-04-10	9500.00	4000.00
8	2026-04-22	4000.00	18000.00
9	2026-05-08	18000.00	6800.00
10	2026-05-20	6800.00	NULL

```sql
-- Display all doctors, including doctors who have no patients assigned to them.
SELECT d.doctor_name, d.specialization,p.patient_name
FROM Doctors d
LEFT JOIN Patients p
ON d.doctor_id = p.doctor_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

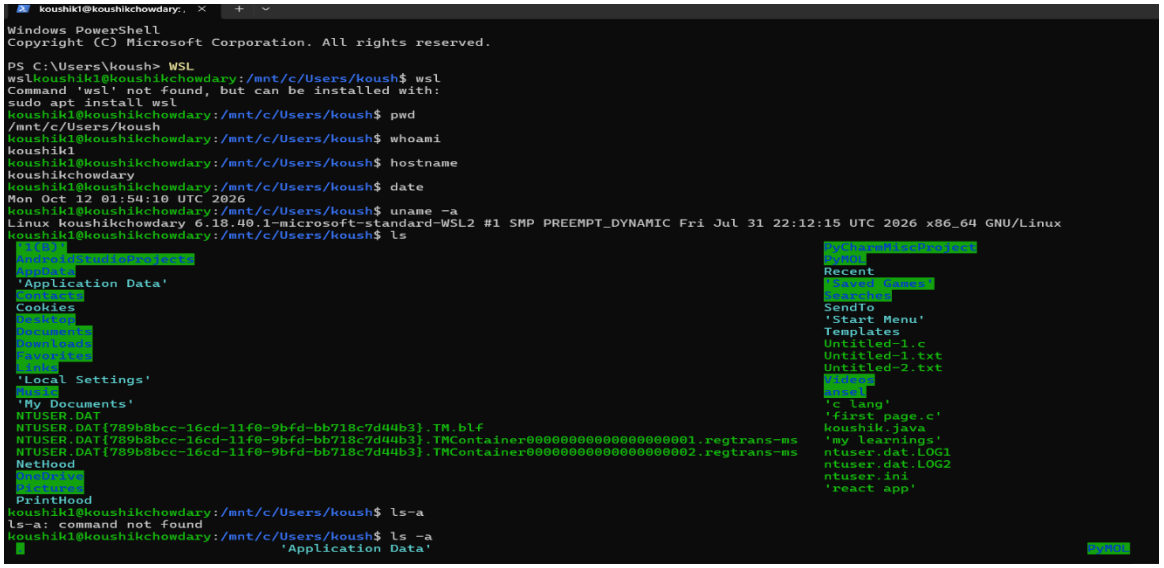
doctor_name	specialization	patient_name
Dr. Anil Kumar	Cardiology	Sneha
Dr. Anil Kumar	Cardiology	Aarav
Dr. Priya Sharma	Dermatology	Divya
Dr. Priya Sharma	Dermatology	Priya
Dr. Ravi Patel	Neurology	Ananya
Dr. Ravi Patel	Neurology	Rahul
Dr. Sneha Reddy	Orthopedics	Vikram
Dr. Sneha Reddy	Orthopedics	Arjun
Dr. Kiran Rao	Pediatrics	Meera
Dr. Kiran Rao	Pediatrics	Kiran
Dr. Meera Nair	Gynecology	NULL
Dr. Arjun Singh	Cardiology	NULL
Dr. Divya Menon	Dermatology	NULL
Dr. Vikram Das	Neurology	NULL
Dr. Ananya Gupta	Orthopedics	NULL

```sql
-- Count the number of patients assigned to each doctor.
SELECT d.doctor_name, COUNT(p.patient_id) AS total_patients
FROM Doctors d
LEFT JOIN Patients p
ON d.doctor_id = p.doctor_id
GROUP BY d.doctor_id, d.doctor_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

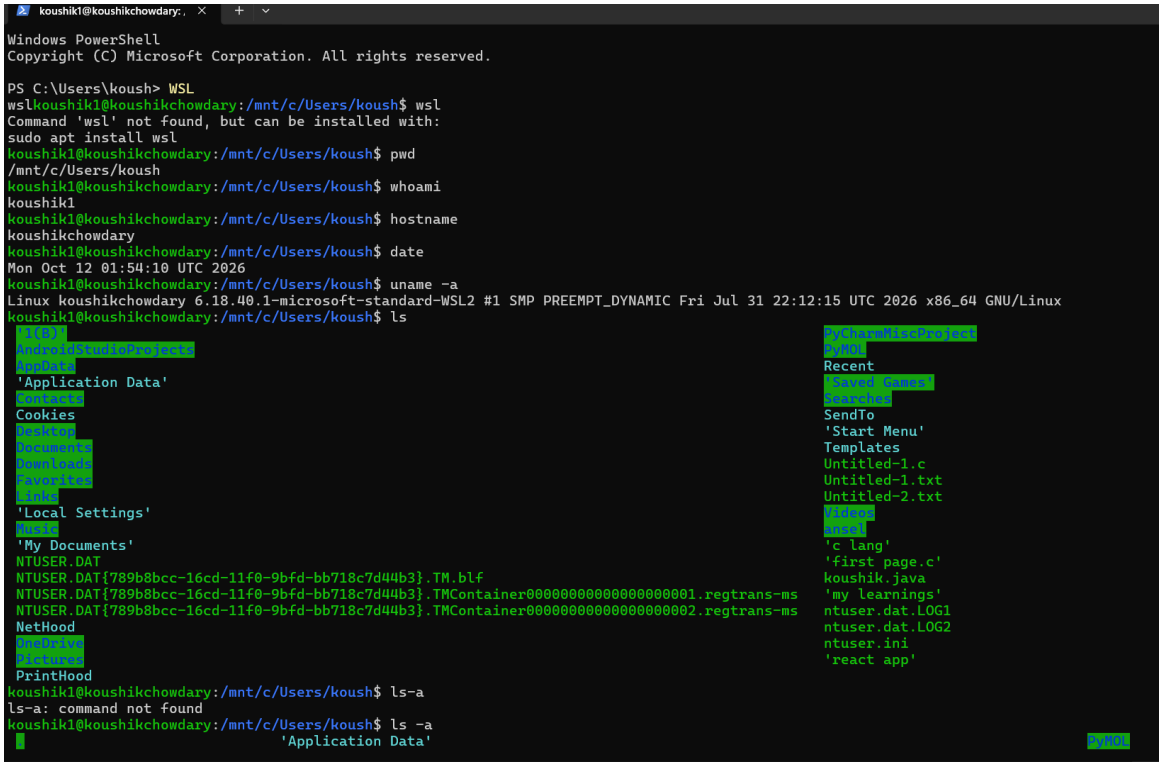
doctor_name	total_patients
Dr. Anil Kumar	2
Dr. Priya Sharma	2
Dr. Ravi Patel	2
Dr. Sneha Reddy	2
Dr. Kiran Rao	2
Dr. Meera Nair	0
Dr. Arjun Singh	0
Dr. Divya Menon	0
Dr. Vikram Das	0
Dr. Ananya Gupta	0

Linux Commands:



```
koushik1@koushikchowdary:.
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\koush> WSL
wslkoushik1@koushikchowdary:/mnt/c/Users/koush$ wsl
Command 'wsl' not found, but can be installed with:
sudo apt install wsl
koushik1@koushikchowdary:/mnt/c/Users/koush$ pwd
/mnt/c/Users/koush
koushik1@koushikchowdary:/mnt/c/Users/koush$ whoami
koushik1
koushik1@koushikchowdary:/mnt/c/Users/koush$ hostname
koushikchowdary
koushik1@koushikchowdary:/mnt/c/Users/koush$ date
Mon Oct 12 01:54:10 UTC 2026
koushik1@koushikchowdary:/mnt/c/Users/koush$ uname -a
Linux koushikchowdary 6.18.40.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Fri Jul 31 22:12:15 UTC 2026 x86_64 GNU/Linux
koushik1@koushikchowdary:/mnt/c/Users/koush$ ls
 '$(@)'                                                                                  PyCharmMiscProject
 AndroidStudioProjects                                                                   PyMOL
 AppData                                                                                 Recent
 'Application Data'                                                                      'Saved Games'
 Contacts                                                                                Searches
 Cookies                                                                                 SendTo
 Desktop                                                                                 'Start Menu'
 Documents                                                                               Templates
 Downloads                                                                               Untitled-1.c
 Favorites                                                                               Untitled-1.txt
 Links                                                                                   Untitled-2.txt
 'Local Settings'                                                                        Videos
 Music                                                                                   ansel
 'My Documents'                                                                          'c lang'
 NTUSER.DAT                                                                              'first page.c'
 NTUSER.DAT{789b8bcc-16cd-11f0-9bfd-bb718c7d44b3}.TM.blf                                 koushik.java
 NTUSER.DAT{789b8bcc-16cd-11f0-9bfd-bb718c7d44b3}.TMContainer00000000000000000001.regtrans-ms   'my learnings'
 NTUSER.DAT{789b8bcc-16cd-11f0-9bfd-bb718c7d44b3}.TMContainer00000000000000000002.regtrans-ms   ntuser.dat.LOG1
 NetHood                                                                                 ntuser.dat.LOG2
 OneDrive                                                                                ntuser.ini
 Pictures                                                                                'react app'
 PrintHood
koushik1@koushikchowdary:/mnt/c/Users/koush$ ls-a
ls-a: command not found
koushik1@koushikchowdary:/mnt/c/Users/koush$ ls -a
                                'Application Data'                                                                      PyMOL
```



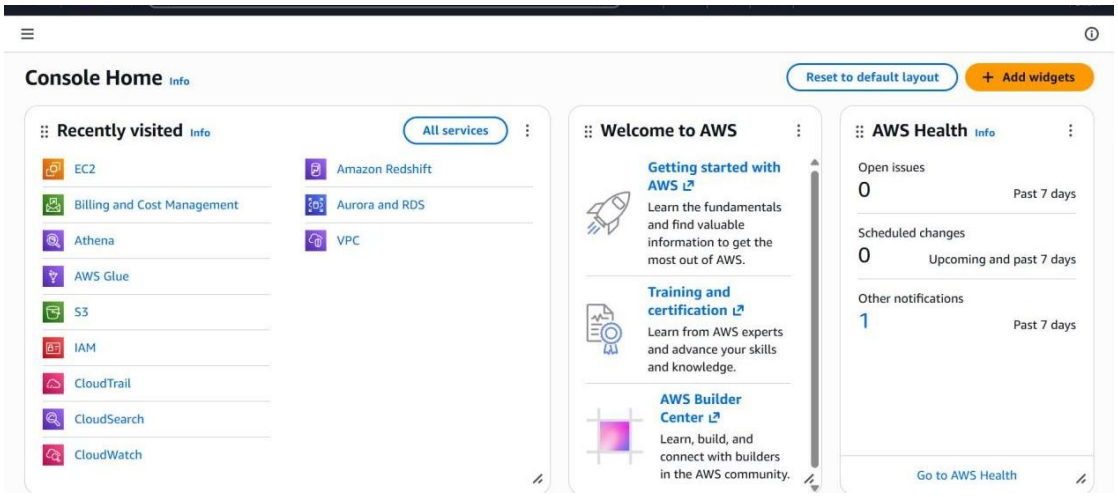

```
koushik1@koushikchowdary:.
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\koush> WSL
wslkoushik1@koushikchowdary:/mnt/c/Users/koush$ wsl
Command 'wsl' not found, but can be installed with:
sudo apt install wsl
koushik1@koushikchowdary:/mnt/c/Users/koush$ pwd
/mnt/c/Users/koush
koushik1@koushikchowdary:/mnt/c/Users/koush$ whoami
koushik1
koushik1@koushikchowdary:/mnt/c/Users/koush$ hostname
koushikchowdary
koushik1@koushikchowdary:/mnt/c/Users/koush$ date
Mon Oct 12 01:54:10 UTC 2026
koushik1@koushikchowdary:/mnt/c/Users/koush$ uname -a
Linux koushikchowdary 6.18.40.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Fri Jul 31 22:12:15 UTC 2026 x86_64 GNU/Linux
koushik1@koushikchowdary:/mnt/c/Users/koush$ ls
 '$(@)'                                                                                  PyCharmMiscProject
 AndroidStudioProjects                                                                   PyMOL
 AppData                                                                                 Recent
 'Application Data'                                                                      'Saved Games'
 Contacts                                                                                Searches
 Cookies                                                                                 SendTo
 Desktop                                                                                 'Start Menu'
 Documents                                                                               Templates
 Downloads                                                                               Untitled-1.c
 Favorites                                                                               Untitled-1.txt
 Links                                                                                   Untitled-2.txt
 'Local Settings'                                                                        Videos
 Music                                                                                   ansel
 'My Documents'                                                                          'c lang'
 NTUSER.DAT                                                                              'first page.c'
 NTUSER.DAT{789b8bcc-16cd-11f0-9bfd-bb718c7d44b3}.TM.blf                                 koushik.java
 NTUSER.DAT{789b8bcc-16cd-11f0-9bfd-bb718c7d44b3}.TMContainer00000000000000000001.regtrans-ms   'my learnings'
 NTUSER.DAT{789b8bcc-16cd-11f0-9bfd-bb718c7d44b3}.TMContainer00000000000000000002.regtrans-ms   ntuser.dat.LOG1
 NetHood                                                                                 ntuser.dat.LOG2
 OneDrive                                                                                ntuser.ini
 Pictures                                                                                'react app'
 PrintHood
koushik1@koushikchowdary:/mnt/c/Users/koush$ ls-a
ls-a: command not found
koushik1@koushikchowdary:/mnt/c/Users/koush$ ls -a
                                'Application Data'                                                                      PyMOL
```

```
14  ls
15  cd linux-practice
16  pwd
17  cd __
18  cd ..
19  pwd
20  mkdir projects
21  mkdir documents
22  mkdir practice
23  ls
24  touch hello.txt
25  ls
26  touch file1.text file2.text file3.text
27  ls
28  echo "Hello linux" > hello.text
29  cat hello.txt
30  cat hello.text
31  echo "Second" >> hello.text
32  cat hello.text
33  touch original.txt
34  cp original.txt copy.txt
35  ls
36  pwd
37  ls
38  ls -l
39  ls -a
40  whoami
41  date
42  cal
43  mkdir linux_practice
44  touch file1.txt
45  echo "Hello Linux"
46  echo "Linux Practice" > file1.txt
47  cat file1.txt
48  cp file1.txt file2.txt
49  mv file1.txt renamed.txt
50  wc -l file1.txt
51  touch file1.txt
```

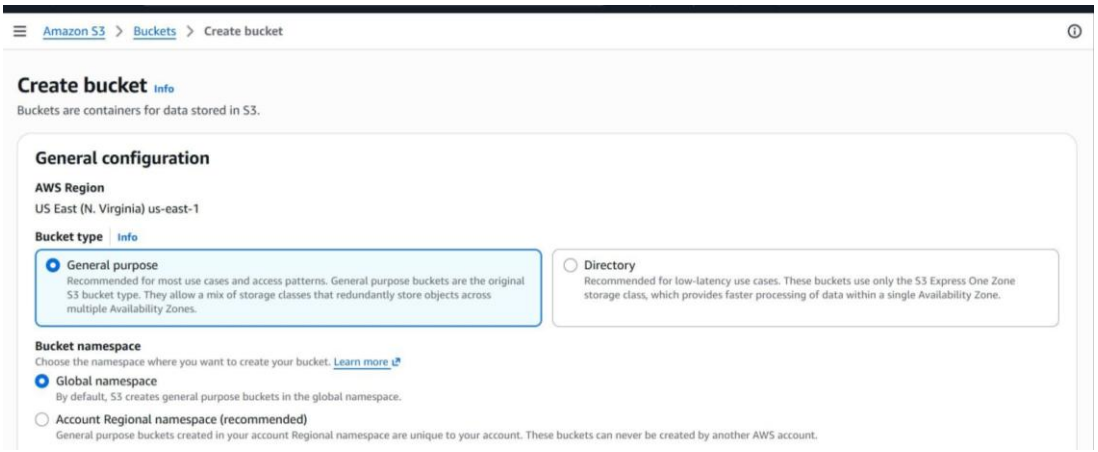
```
55  cat file1.txt
56  wc -l file1.txt
57  head -n 5 file1.txt
58  free -h
59  uname -a
60  uname -a\
61  uname -a
62  df -h
63  wc -w notes.txt
64  clear
65  pwd
66  whoami
67  hostname
68  date
69  uname -a
70  ls
71  ls -a
72  id
73  df -h
74  mkdir assignment
75  cd assignment
76  touch notes.txt
77  echo "Hello Linux" > notes.txt
78  cat notes.txt
79  cp notes.txt backup.txt
80  mv copy.txt backup.txt
81  cp notes.txt copy.txt
82  mv copy.txt backup.txt
83  ls -l
84  wc -w notes.txt
85  wc -l notes.txt
86  history
```

AWS Account / Region:

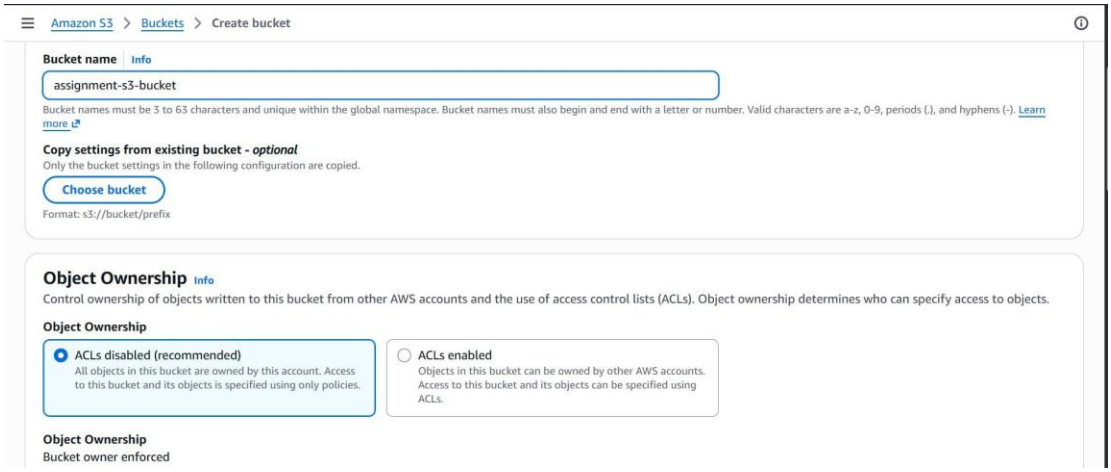


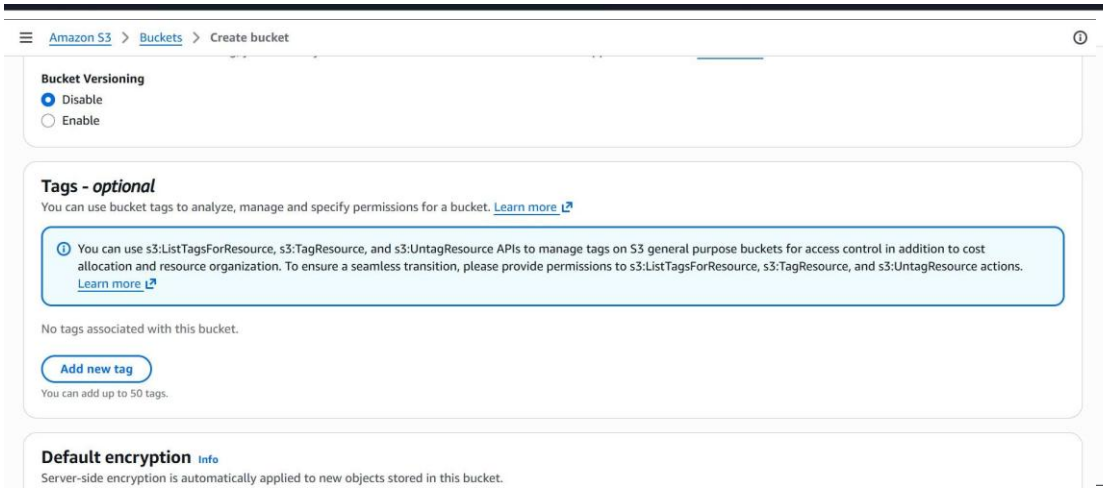
Amazon S3:

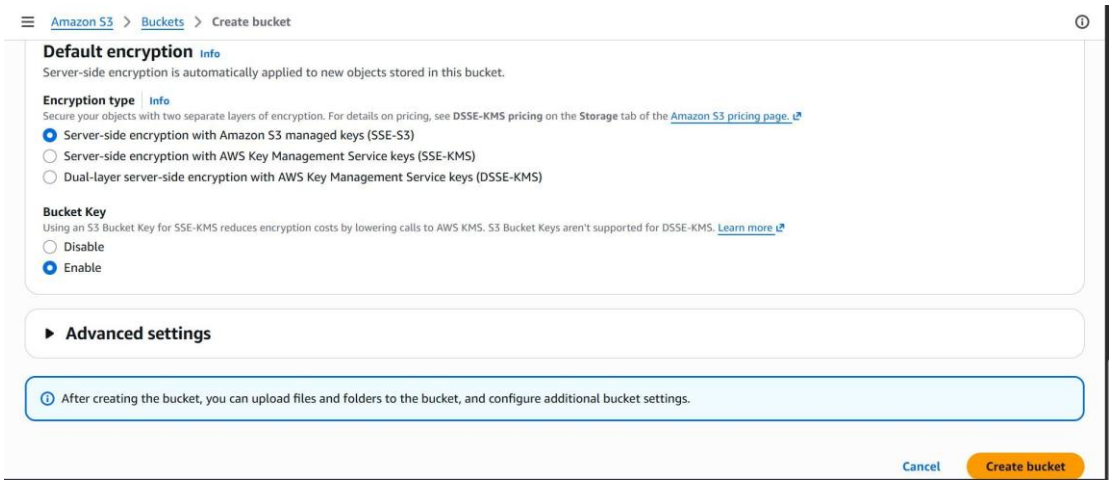
Create an S3 bucket and keep the bucket type as general purpose



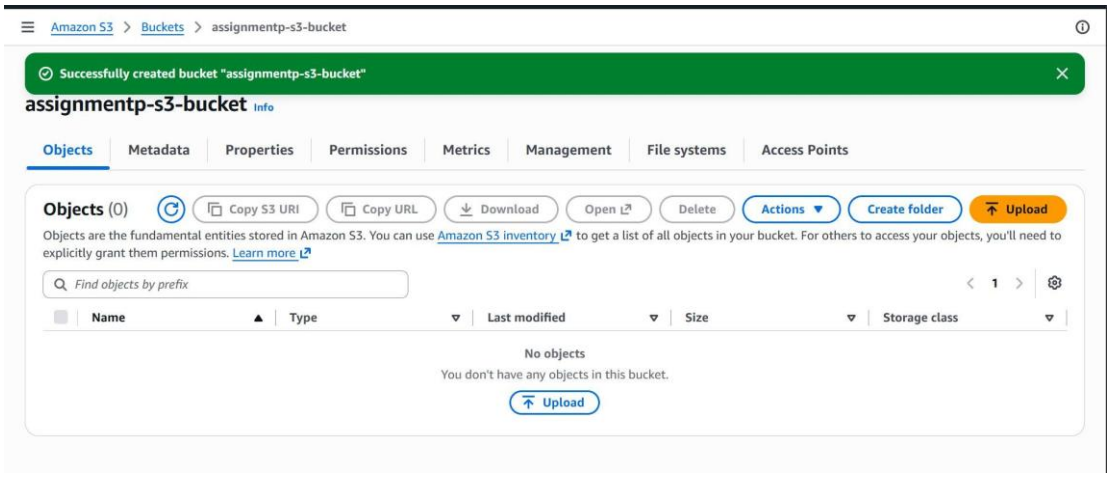
Enter a unique bucket name



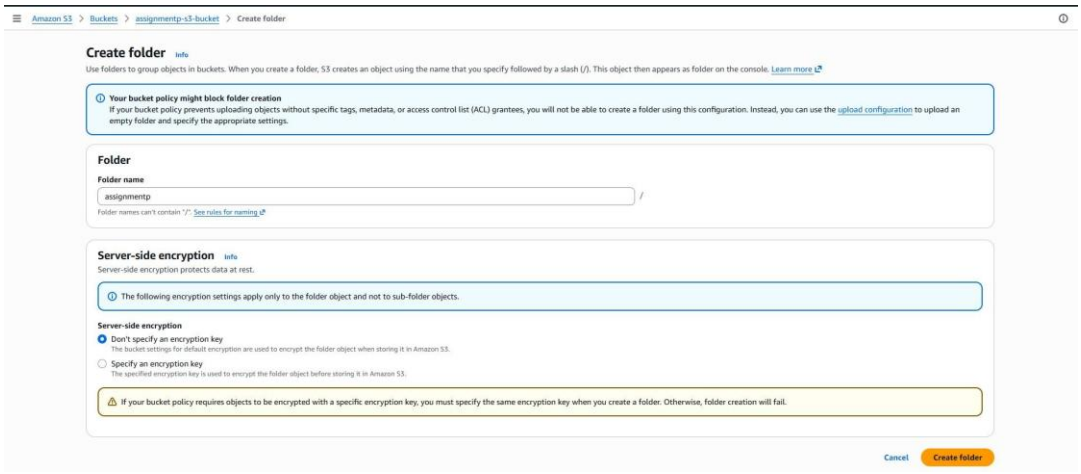


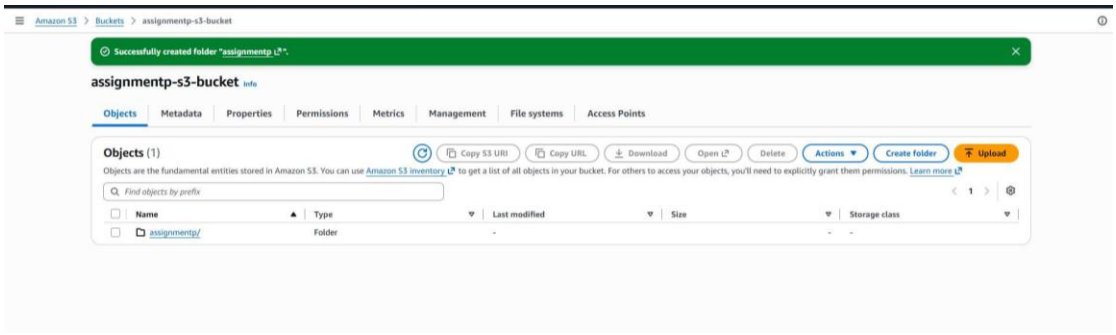


Bucket is created successfully and now appears empty with 0 objects

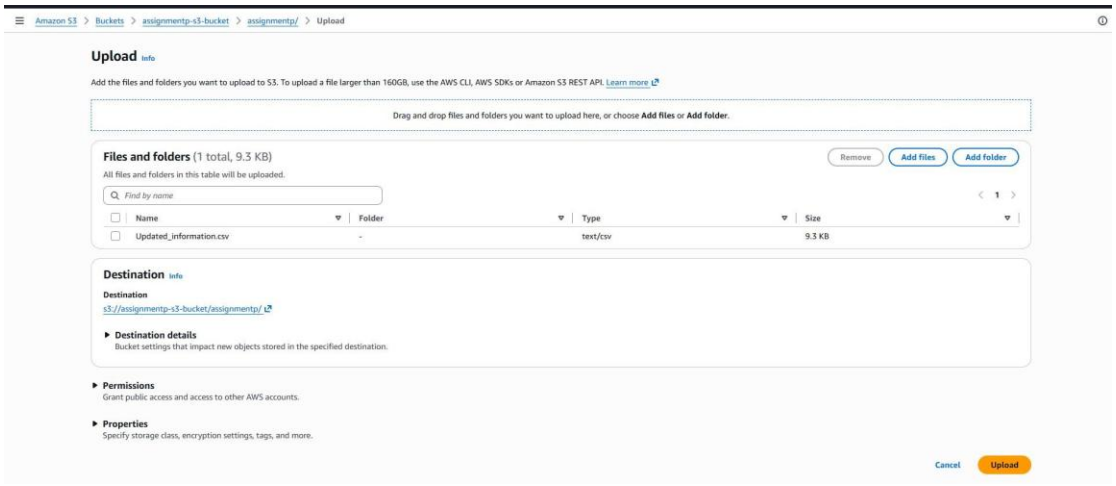


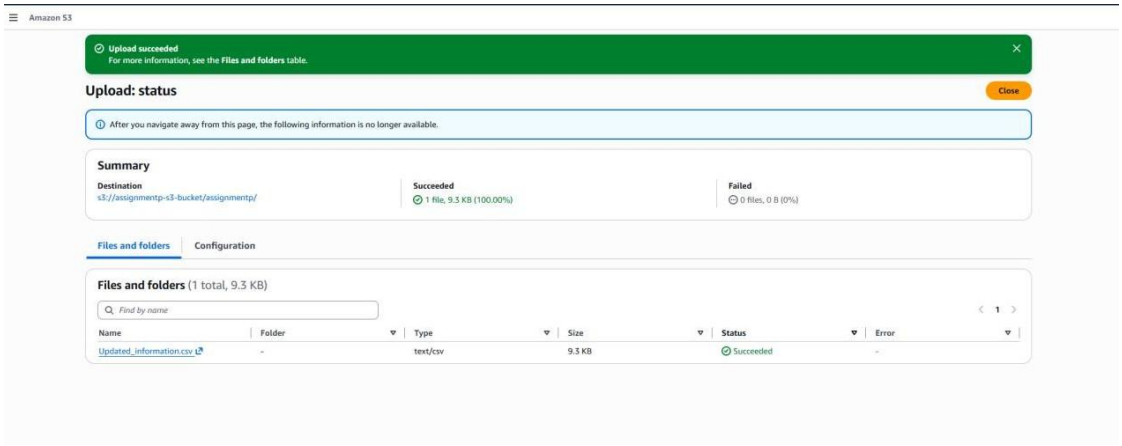
Create a folder

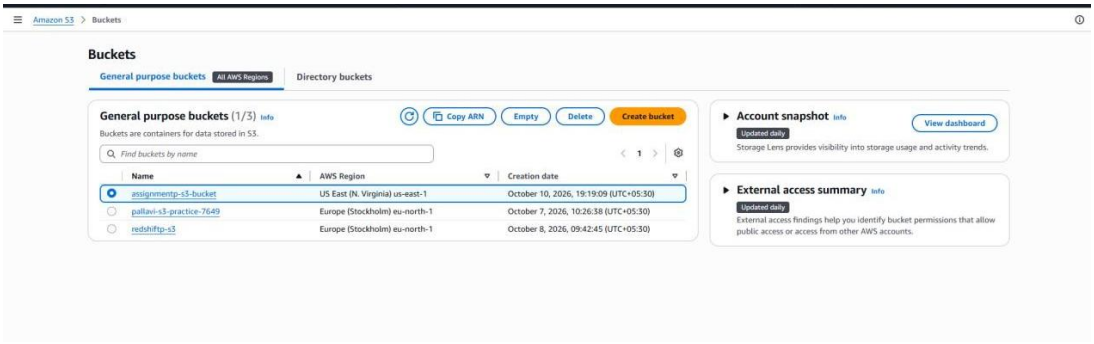




Upload the file

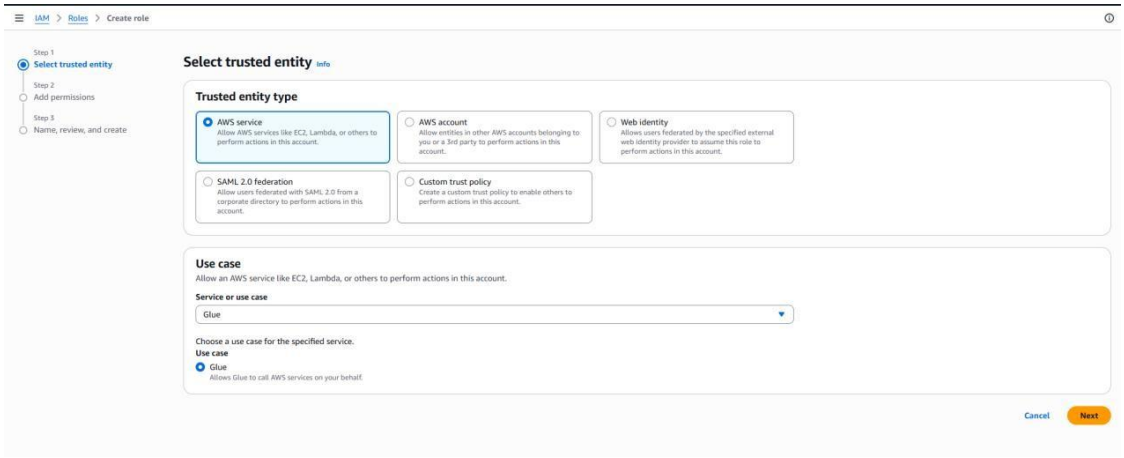




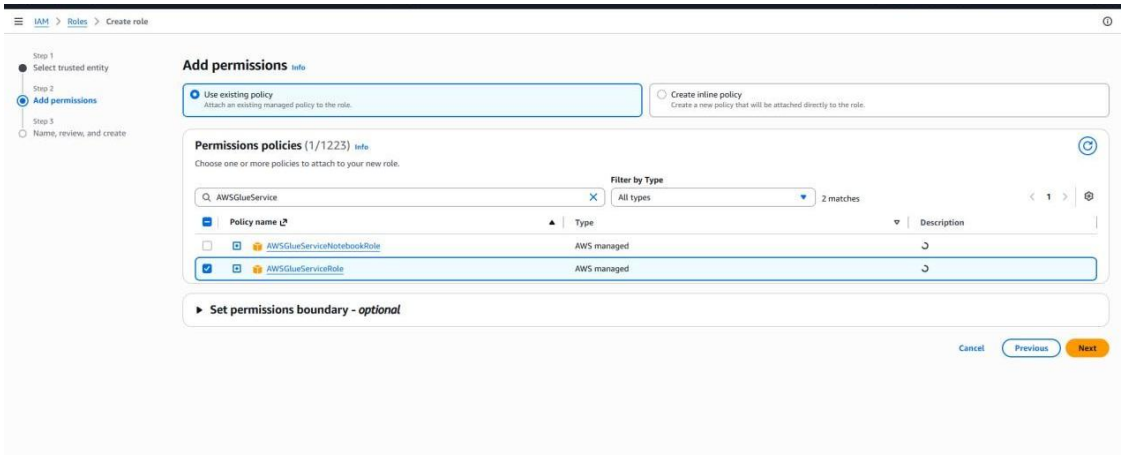


IAM:

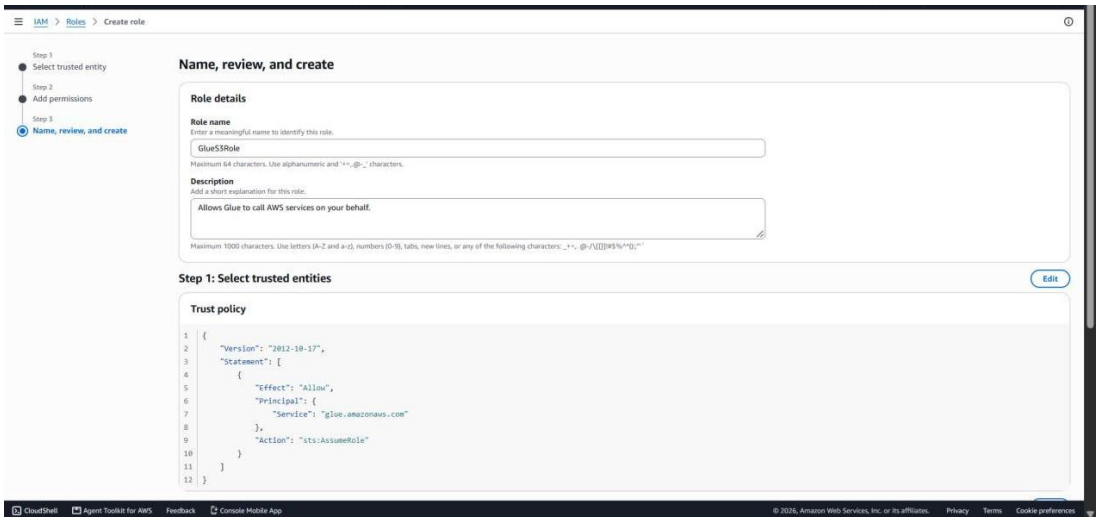
Select trusted entity as AWS service

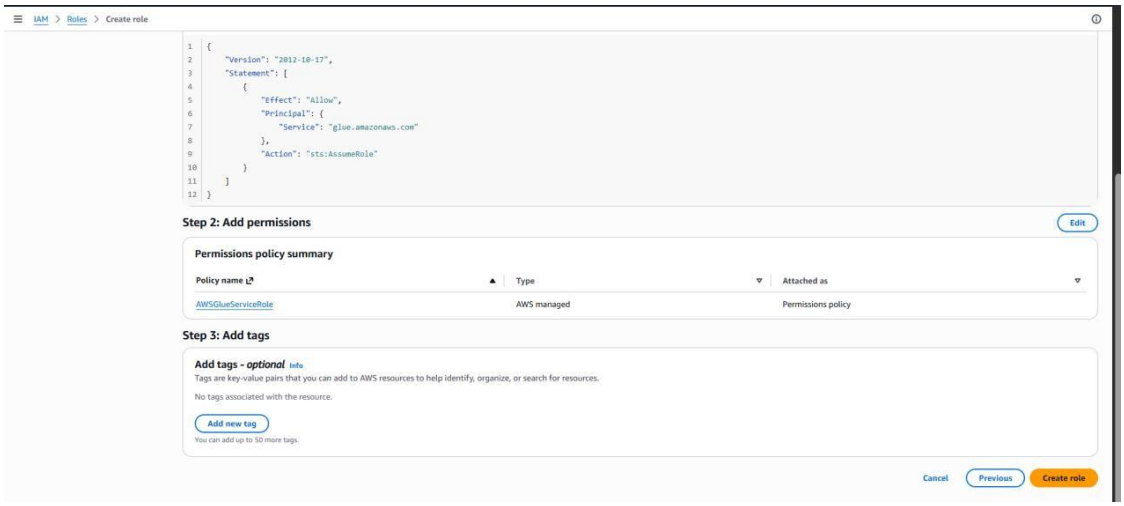


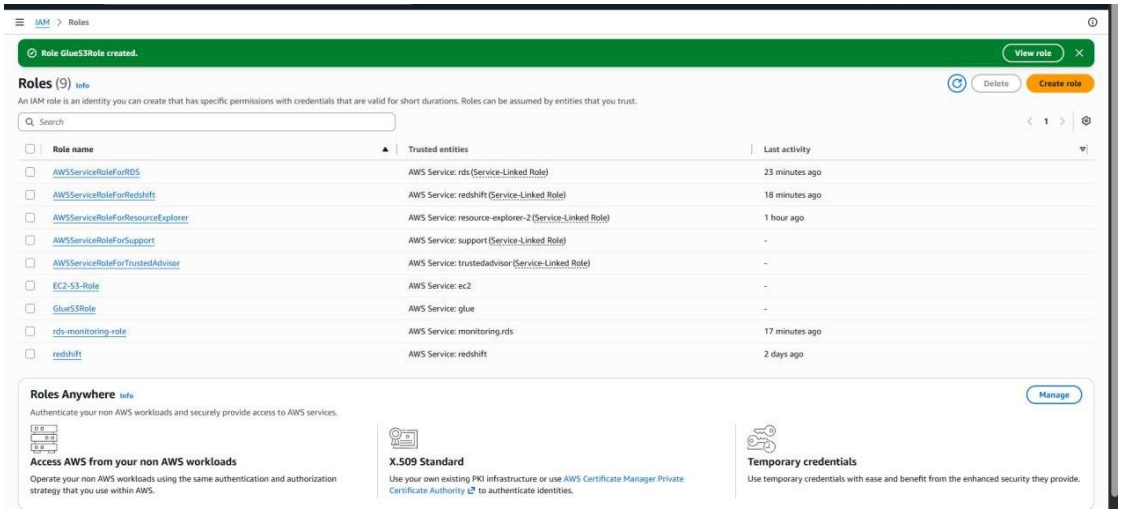
Add permissions



Role name

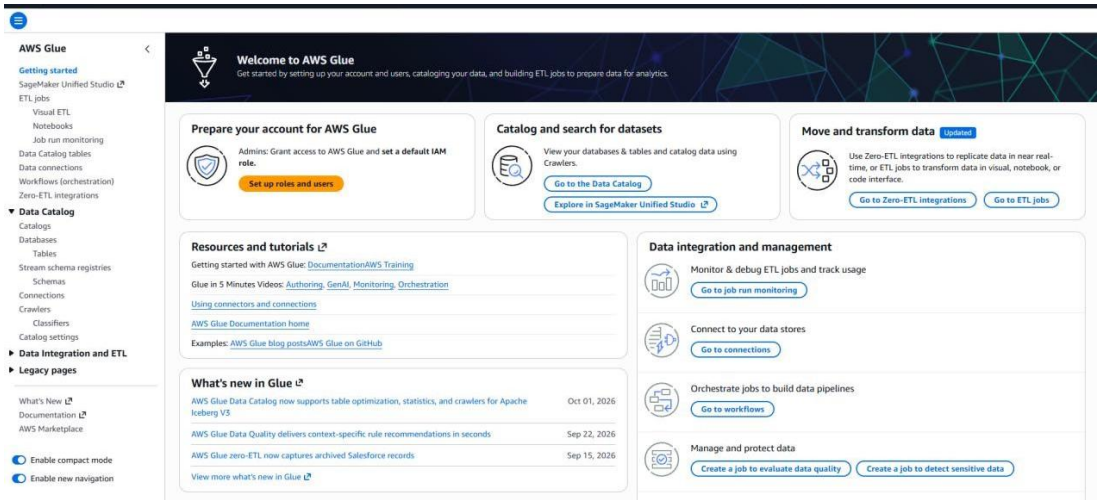




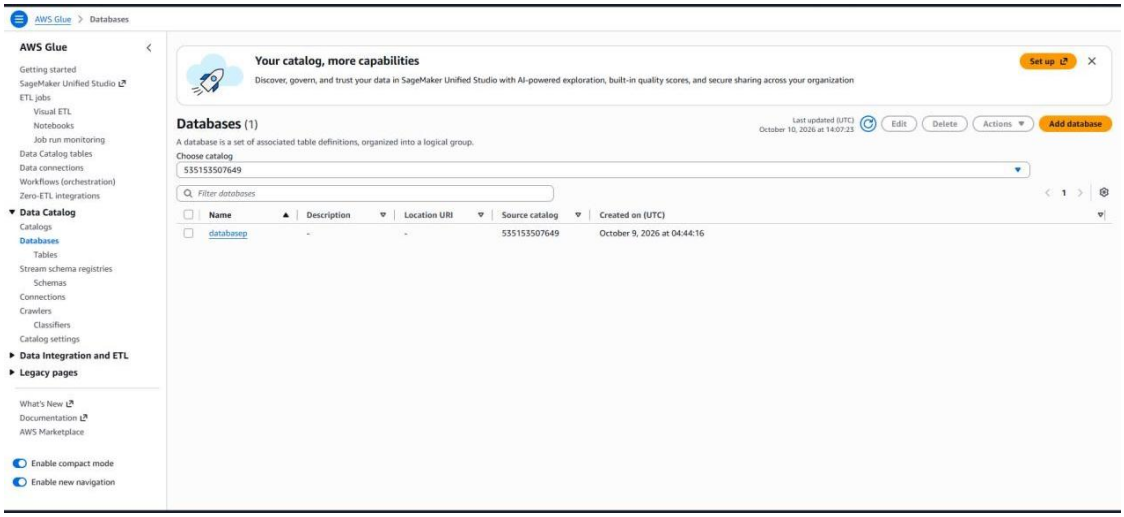


GLUE:

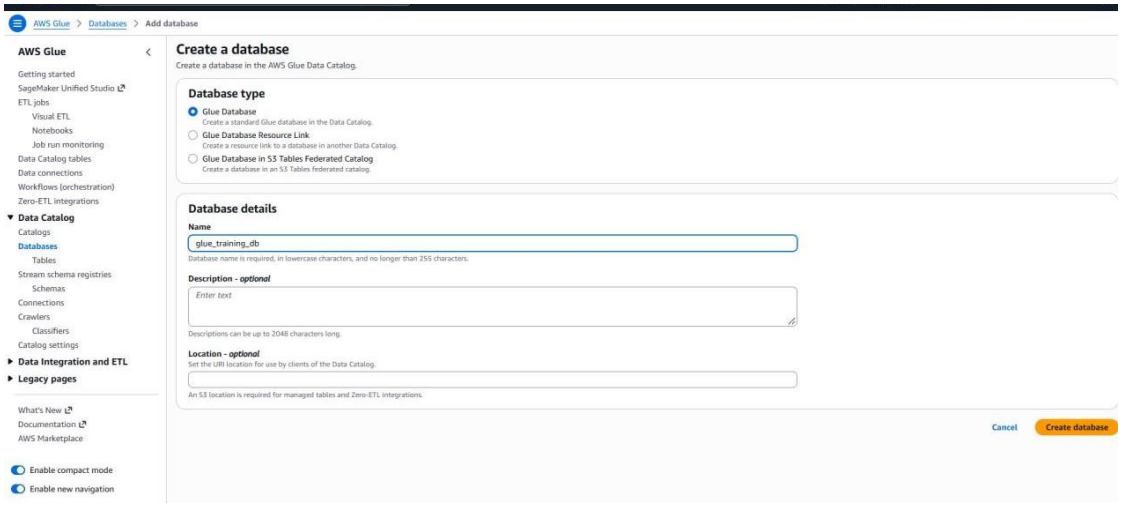
AWS Glue Console



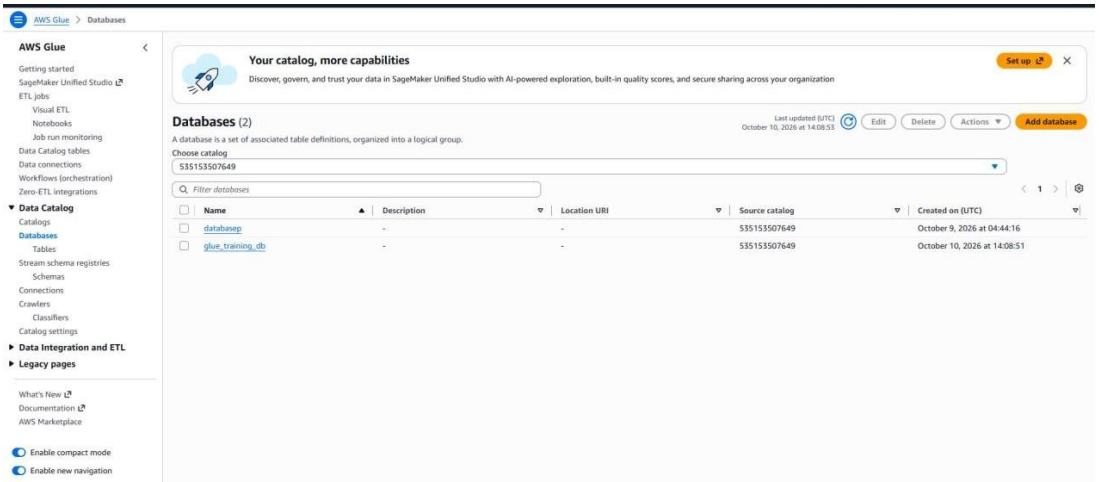
Data Catalog -> Databases



Create a database

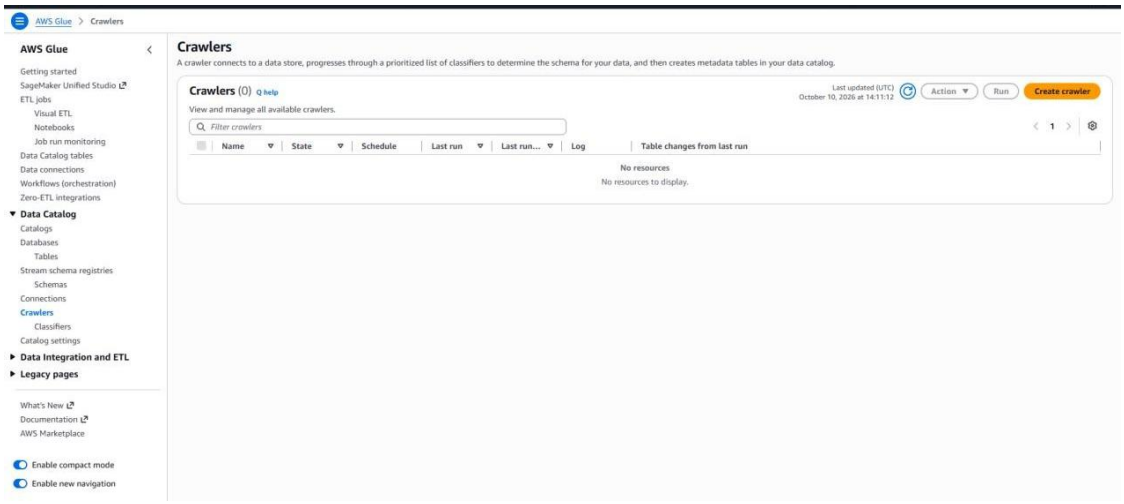


Database list now showing glue_training_db

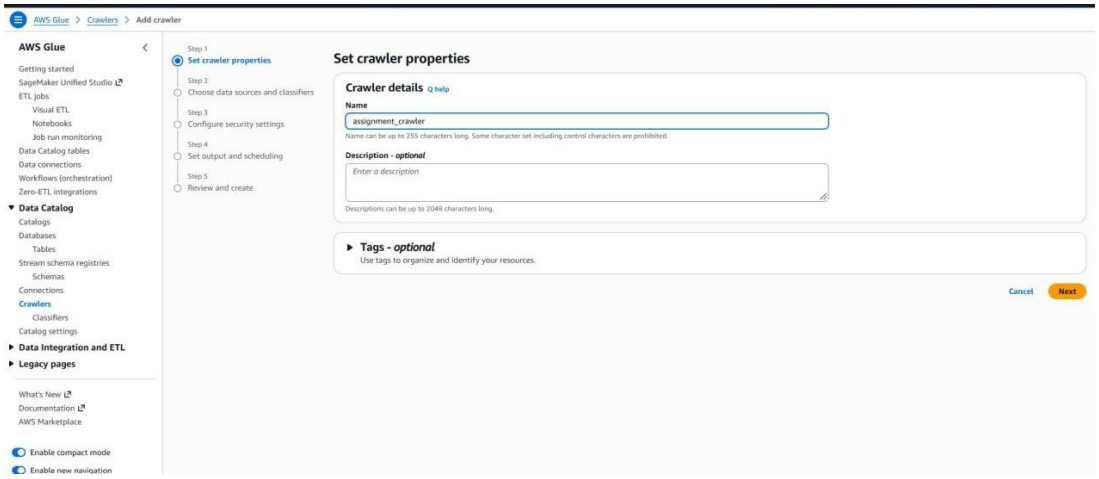


AWS CRAWLER

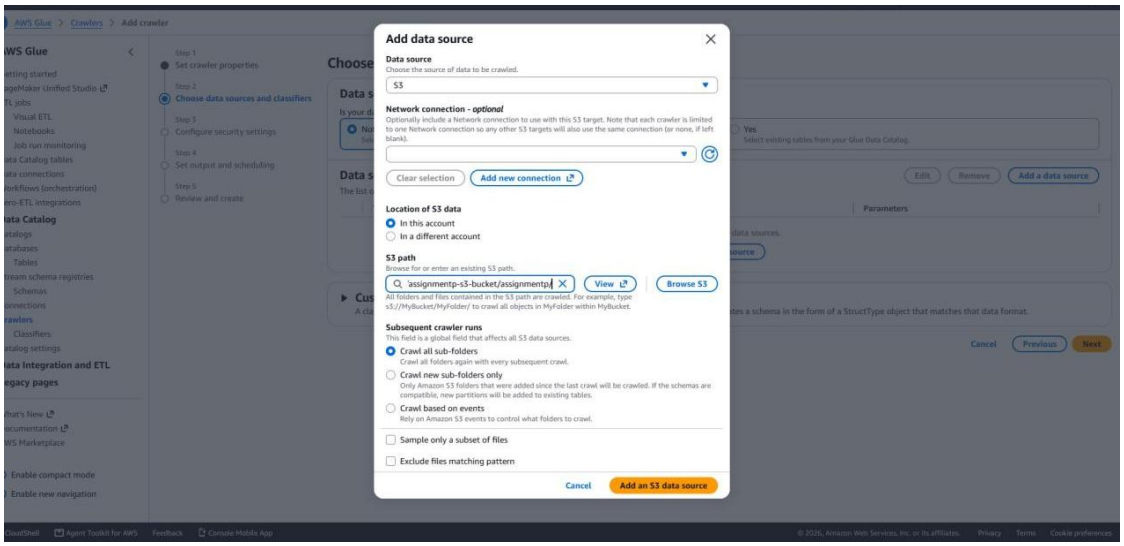
Crawler page with no crawlers yet



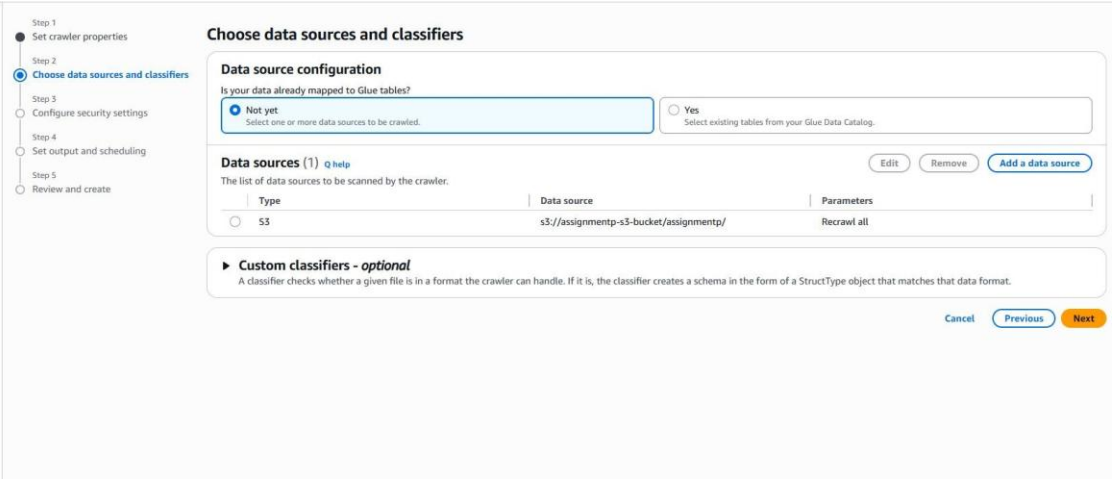
Crawler properties page with the crawler name entered as assignment_crawler.



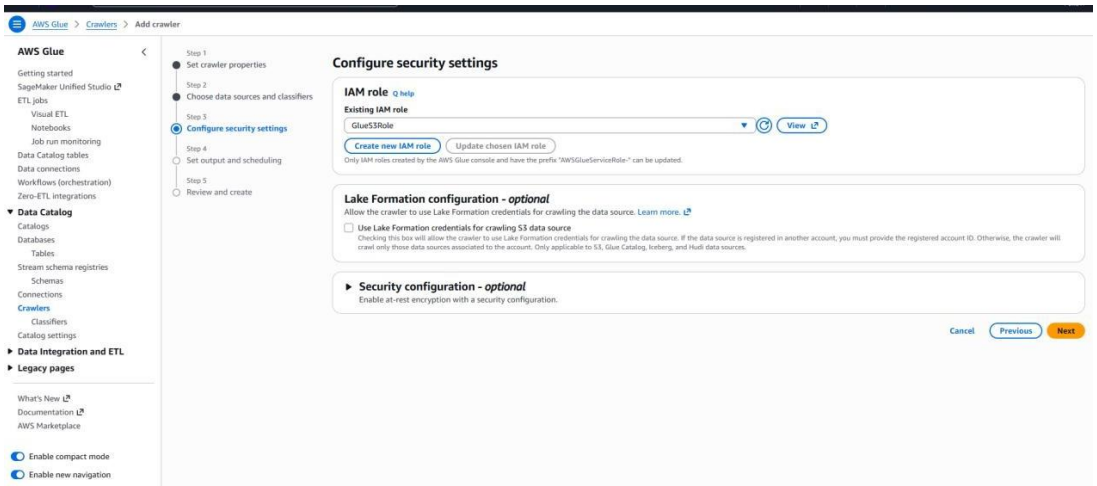
S3 source location



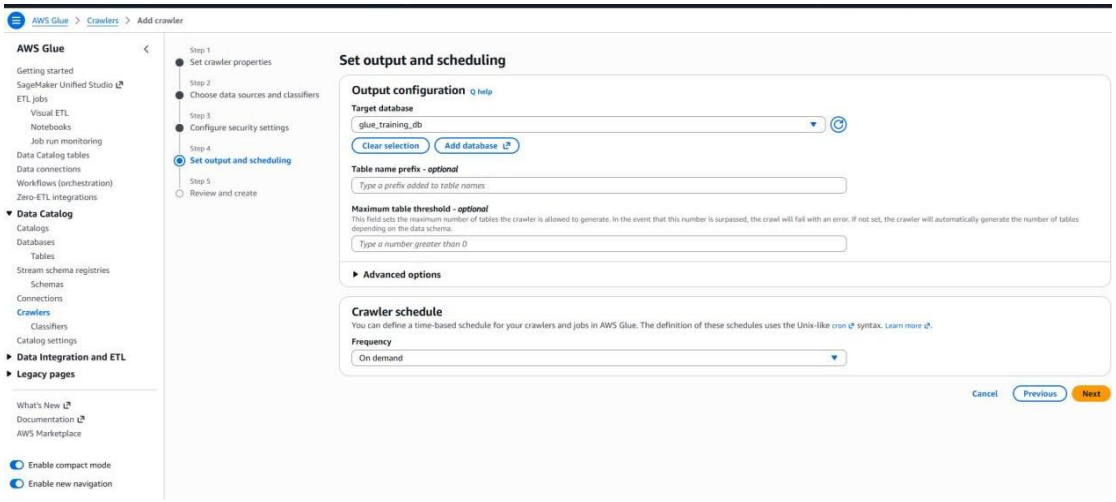
Data Source added



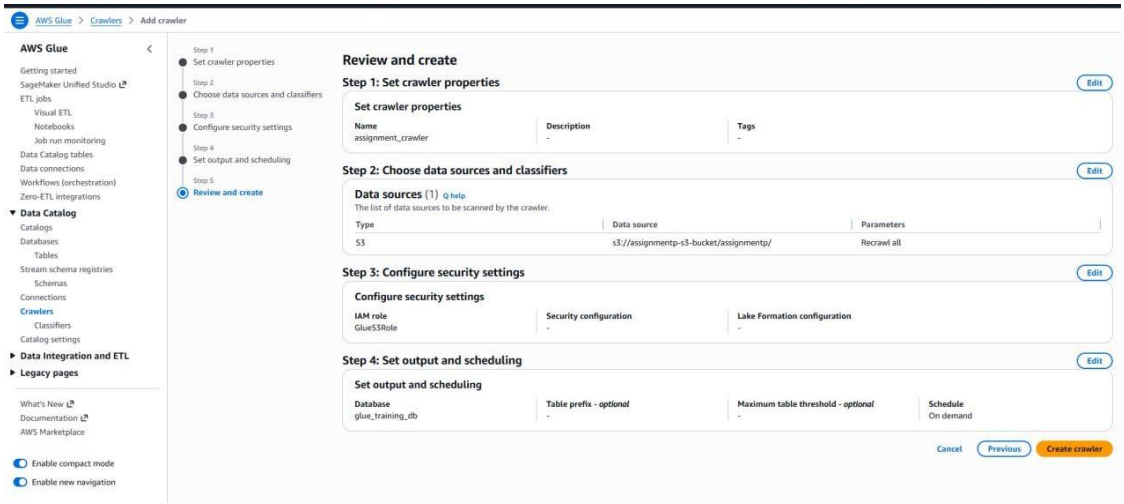
Configure security settings page with IAM role

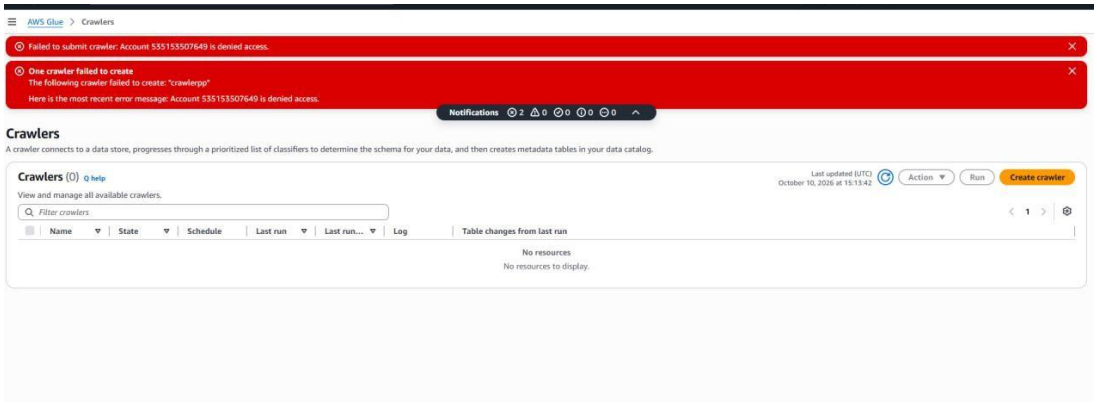


Output and scheduling page with glue_training_db as the target database



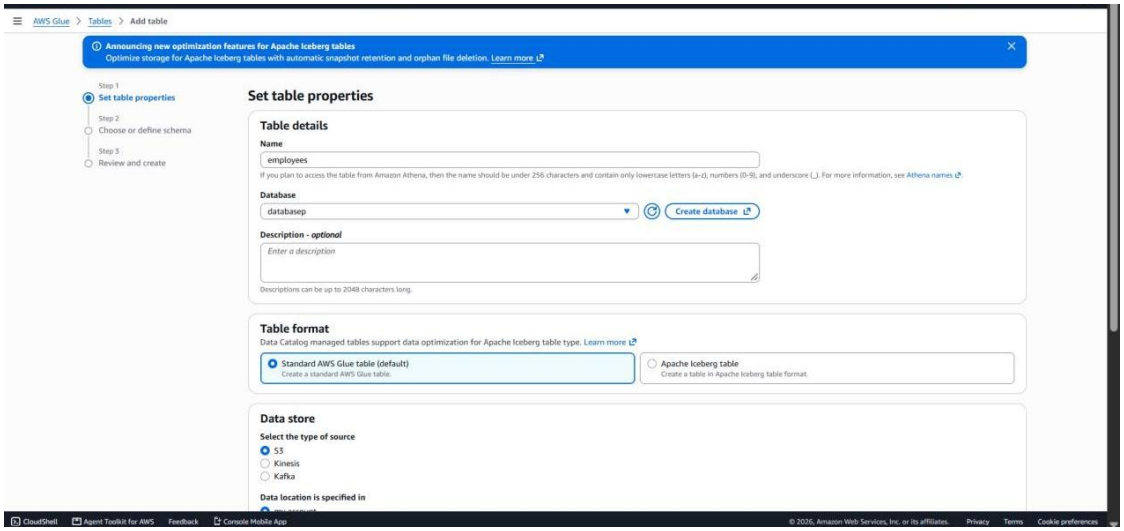
Review and create

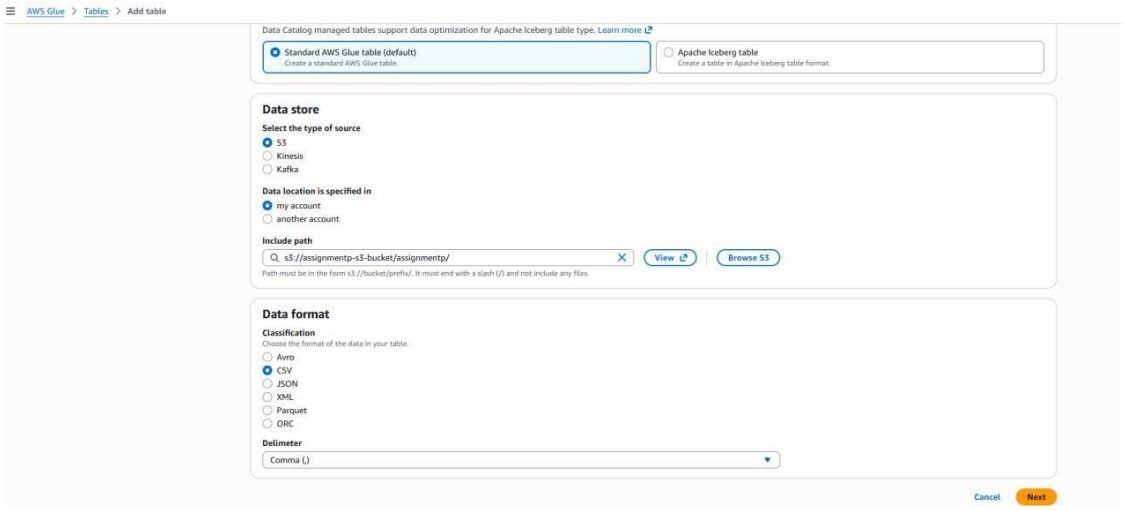


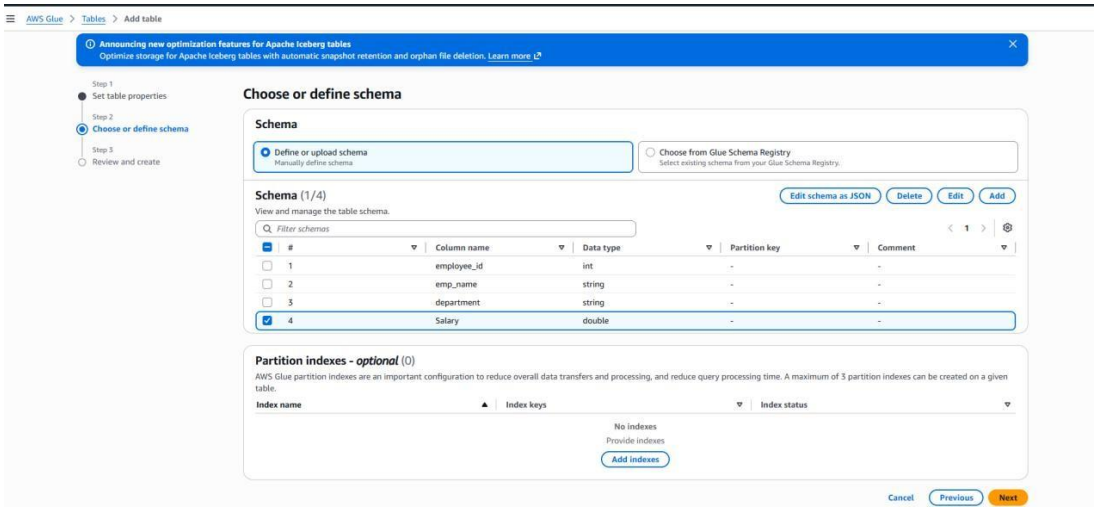


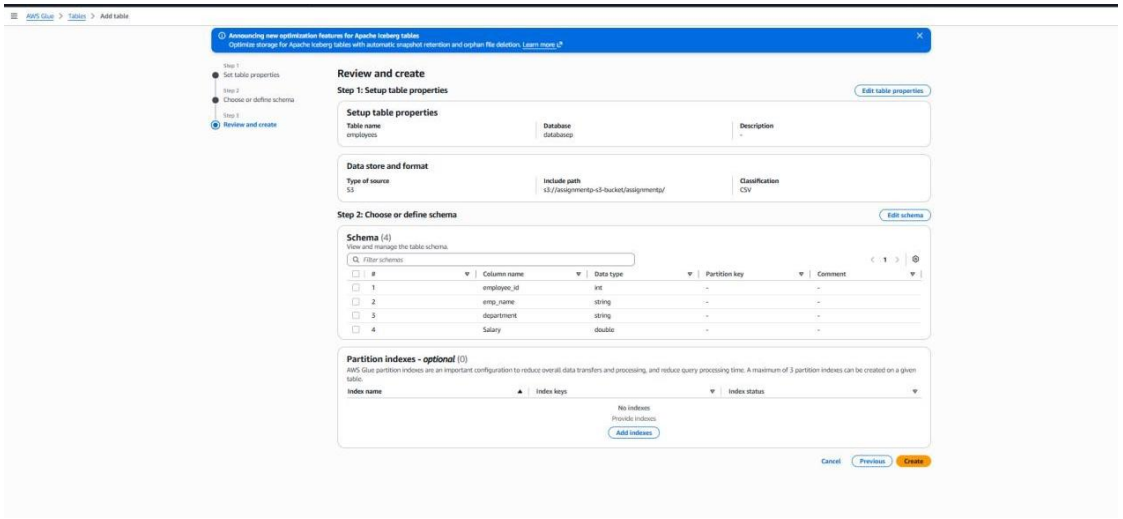
GLUE DATA CATALOG

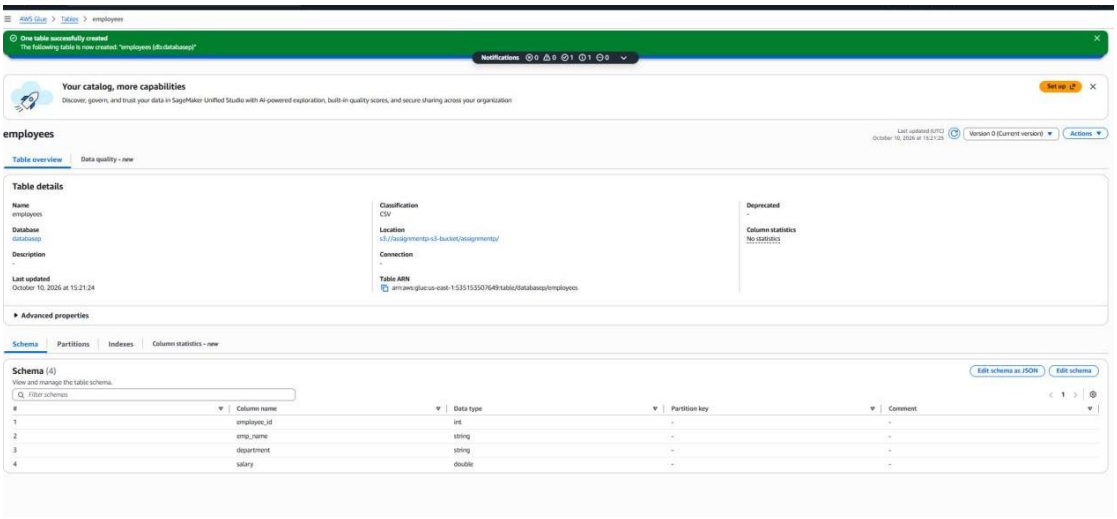
Set table properties



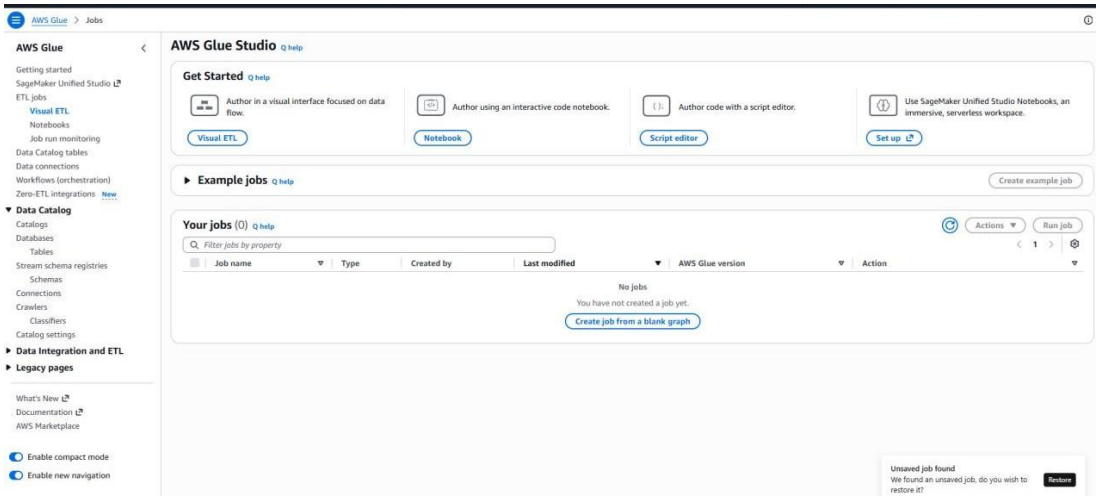


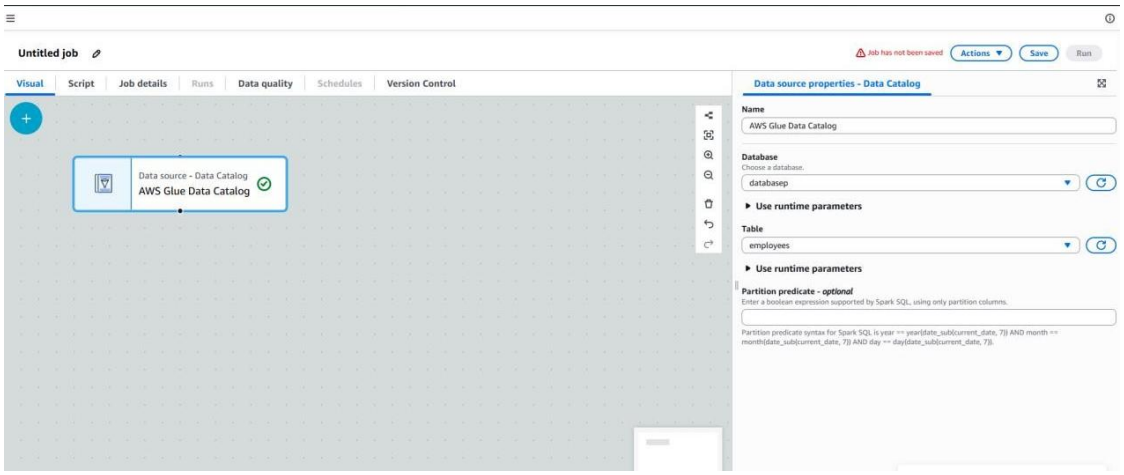


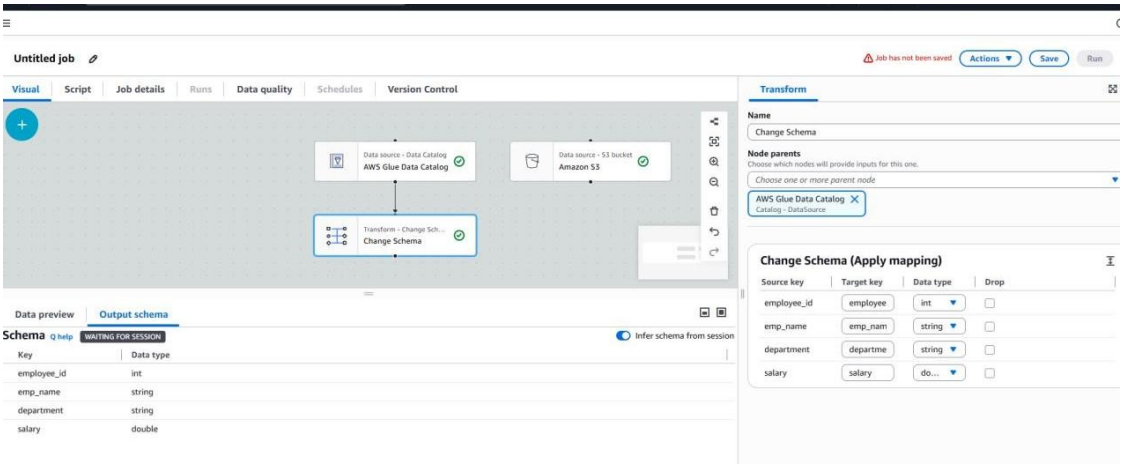


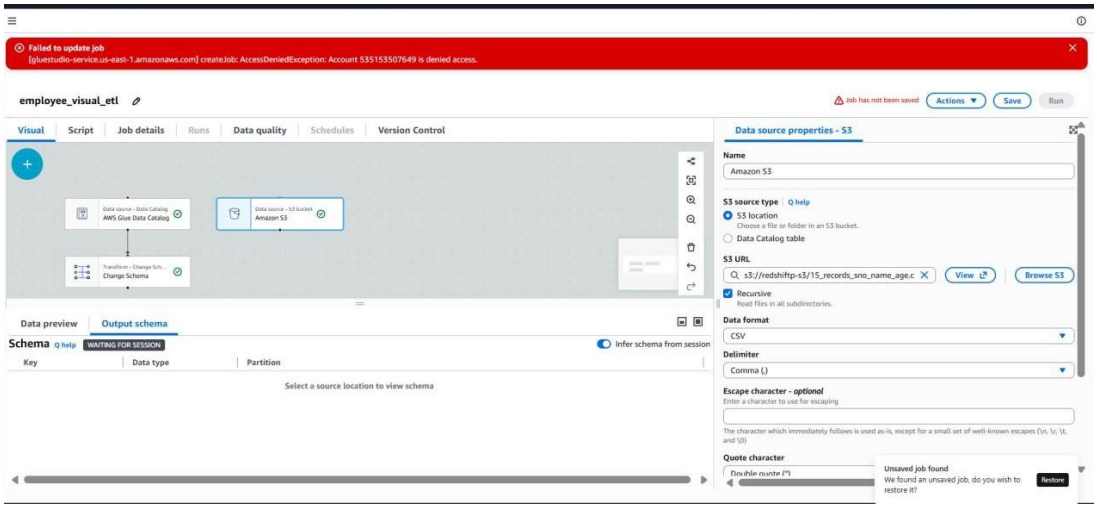


GLUE VISUAL ETL

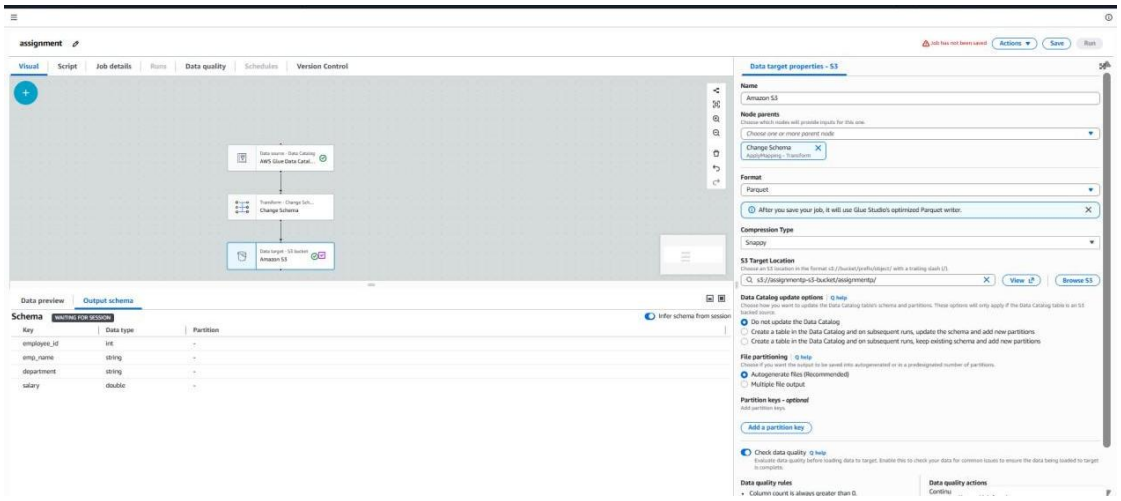


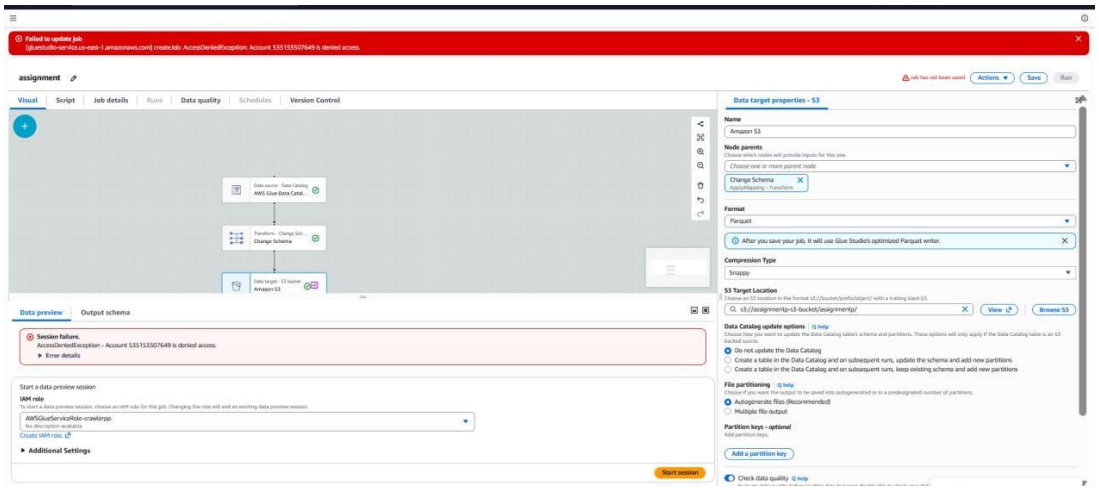

Untitled job
Job has not been saved
Actions
Save
Run
Visual
Script
Job details
Runs
Data quality
Schedules
Version Control
Data source - Data Catalog
AWS Glue Data Catalog
Data source properties - Data Catalog
Name
AWS Glue Data Catalog
Database
Choose a database.
databasep
Use runtime parameters
Table
employees
Use runtime parameters
Partition predicate - optional
Enter a boolean expression supported by Spark SQL, using only partition columns.


Untitled job
Job has not been saved
Actions
Save
Run
Visual
Script
Job details
Runs
Data quality
Schedules
Version Control
Data source - Data Catalog
AWS Glue Data Catalog
Data source - S3 bucket
Amazon S3
Transform - Change Sch...
Change Schema
Transform
Name
Change Schema
Node parents
Choose which nodes will provide inputs for this one.
Choose one or more parent node
AWS Glue Data Catalog
Catalog - DataSource
Change Schema (Apply mapping)
Source key
Target key
Data type
Drop
employee_id
employee
int
emp_name
emp_nam
string
department
departme
string
salary
salary
do...
Data preview
Output schema
Schema
Q help
WAITING FOR SESSION
Infer schema from session
Key
Data type
employee_id
int
emp_name
string
department
string
salary
double


Failed to update job
[gluestudio-service.us-east-1.amazonaws.com] createJob: AccessDeniedException: Account 535153507649 is denied access.
employee_visual_etl
Job has not been saved
Actions
Save
Run
Visual
Script
Job details
Runs
Data quality
Schedules
Version Control
Data source - Data Catalog
AWS Glue Data Catalog
Data source - S3 bucket
Amazon S3
Transform - Change Sch...
Change Schema
Data source properties - S3
Name
Amazon S3
S3 source type
Q help
S3 location
Choose a file or folder in an S3 bucket.
Data Catalog table
S3 URL
s3://redshiftp-s3/15_records_sno_name_age.c
View
Browse S3
Recursive
Read files in all subdirectories.
Data format
CSV
Delimiter
Comma (,)
Escape character - optional
Enter a character to use for escaping
Quote character
Data preview
Output schema
Schema
Q help
WAITING FOR SESSION
Infer schema from session
Key
Data type
Partition
Select a source location to view schema
Unsaved job found
We found an unsaved job, do you wish to restore it?
Restore

S3 Target Location





FINAL EXPLANATION: My project is based on AWS Glue ETL workflow, where I used S3 to store the customer_details.csv dataset and process the data for analytics.

First, I uploaded the customer_details.csv file into the raw folder of my S3 bucket. Then I created a Glue Crawler to scan the dataset and identify the column names, data types and structure of the data.

After that, the crawler stores the metadata in AWS Glue Data Catalog. I created a database named glue_db and a table called customer_details to manage the dataset information.

Next, I used AWS Glue Visual ETL to design the workflow by connecting the source, transformation and target nodes. It helps to process the data without writing the complete code manually.

In the transformation step, I used the Change Schema option to select the required columns, rename columns and modify data types if needed. After completing the transformation, the processed data is stored in the curated folder of my S3 bucket in Parquet format. Parquet is useful for analytics because it stores data in a column-based format.

Finally, this workflow separates the raw data and processed data, which makes the dataset more organized and easy to analyze using services like Amazon Athena.

Through this project, I learned how AWS Glue is used for data discovery, metadata management and ETL processing. The final output will be available in the curated folder after the Glue ETL job runs successfully.